Appendix Materials to

Voice or Veto? Participatory Procedure in Local Government

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Contents

Overview of LLM-Based Measurement

Much of the measurement in this paper is done by large language models (LLMs) reading legal and transcript text at a scale beyond human capabilities. For each measure, I hand-label a random sample—typically stratified, and blind to the model's output—and report standard agreement statistics (percent agreement; Cohen's κ; correlations for continuous scores). Where the estimand is a corpus-level share or mean, I report prediction-powered inference (PPI) estimates (Angelopoulos et al., 2023a,b), which debias the model's aggregate by its measured error on the human-labeled sample and carry the resulting uncertainty into the confidence interval.

1 STARA Survey of the California State Code

1.1 Main Prompt and Validation

Prompt

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TASK: You are extracting structured metadata from a single provision of the California statutory codes. First
     decide whether the provision imposes a public-participation requirement. If it does, code it on every
     field in the schema below.
WHAT COUNTS (relevance gate): A provision is RELEVANT if it imposes on a governmental actor a statutory duty to
      hold, provide, notice, or offer a public-participation opportunity before taking a legally significant
     action, or as part of carrying out a legal duty. A public-participation opportunity is any of: (a) a
     public hearing (oral testimony before a public body), OR (b) a public comment period (written comments
     invited from the public), OR  (c) a public workshop or other public engagement opportunity. The duty must
     run to the PUBLIC, not only to named parties. Treat a provision as public if it (i) says "public hearing
     "/"public comment"/"duly noticed," (ii) requires notice to the general public plus an opportunity to be
     heard before a public body, or (iii) incorporates another section that makes the opportunity public. A
     provision is NOT relevant if it: mentions a hearing without making it public; authorizes participation in
     purely discretionary terms ("may hold a hearing"); is a trial-type or fair-hearing procedure for parties
     only; merely describes conduct "at the hearing" while the requirement itself sits elsewhere; or refers to
     a participation
opportunity arising under another law without itself imposing it.
EVIDENCE AND HONESTY RULES: For every coded field, rely only on the provided provision text (and any text it
     quotes). Do NOT infer from background knowledge of California practice. When the text does not state a
     field, return null (or "not_stated"), never a guess. For each judgment call, populate the paired "_quote"
     field with a short verbatim snippet (<=25 words) from the provision that supports the coding. Use JSON
     arrays for every multi-valued field. Do not concatenate with ";".
CODING NOTES: decision_character: "legislative" = sets general policy/standards applying broadly (plan or
     ordinance adoption, area rezoning, generally applicable rate). "Adjudicative"  = applies existing
     standards to a specific party/parcel/permit/penalty (variance, CUP,  license issuance or revocation,
     individual assessment, sanction). "ministerial" =  non-discretionary application of fixed criteria. Use "
     mixed" only when the same provision plainly does both; otherwise pick the dominant character and explain.
     findings_required vs written_response_required are different duties: findings concern the agency's own
     stated basis for decision; written response concerns replying to public comments. CEQA-style review
     typically has both; a Brown-Act transparency hearing typically has neither. trigger_mechanism (what causes
      the opportunity to occur) is distinct from participant_scope (who may speak).
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Validation While the original paper developing STARA (Surani et al., 2025) provides a broad sense of the tool's performance, I manually validated a random sample of provisions flagged as positive and negative by the tool to gain a rough-sense of context-specific recall (i.e., the proportion of known provisions recovered) and precision (i.e., proportion of provisions that are actually matches). Among 25 positively flagged provisions, I fully agreed with 24 (96%), and the one disagreement was a relatively close call – HSC § 44391.4 mandates a "transparent meaningful public process" but offers no further specifics and hence I tagged it as negative. Among 25 negatively flagged provisions that passed the regular expression filter but then were labeled negative by the LLM, I fully agreed with 22 (88%). On one provision, FAC § 56189.5, I felt it was ambiguous as to whether the "opportunity for hearing" only applied to parties or was possibly open to broader participation. In GOV § 64717, I felt that the provision's incorporation by reference of another method of participation was sufficient to create an independent mandate. GOV. § 19131 specified an "opportunity to comment" that appeared limited to employee organizations and those who request notice, but this felt like it could grow sufficiently large as to constitute public participation.

1.2 Dating the Enactment of Hearing Mandates

The goal is to estimate when each provision–a section of the California Code–first entered law. The parenthetical appended to every section on the Legislature's official site (leginfo.legislature.ca.gov)—e.g., "(Amended by Stats. 2001, Ch. 56.)"—records only the *most recent* action taken on the section. For 965 of the 2,240 provisions (43%), the timestamp happens to be a birth record: the note's verb is "Added," "Enacted," or "Repealed and added," meaning the section has not been touched since its text entered the code. These provisions are dated exactly.

The remaining 1,234 provisions carry only an "Amended by" note, and for them I construct bounds from two facts.[1] First, a section must exist before it can be amended, so the last-amendment year is an upper bound. Second, every section sits inside a statutory container—an article or chapter—whose own enactment note I parse from the citation's hierarchy on the same

[1]41 rows are not imputable because they could not be dated.

site; a section cannot join a container before the container exists, so the container's enactment year is a lower bound. Take, for example, Government Code § 6586.5, whose note reads "Amended by Stats. 2001": its article was added in 1985, so its enactment lies in [1985, 2001]. The intervals are valid but wide—median span 24 years, only 15% within a single decade—so bounds alone cannot assign most censored provisions even to a decade bin.

I impute within the intervals using the exactly dated provisions as a calibration sample. For the 940 with dated containers, the lag between container and section enactment is fully observed: 63% of sections were born with their container (lag zero), the mean lag is 11 years, and the 90th percentile is 42. Each censored provision then draws a lag from exactly dated donor sections whose containers were enacted within ±10 years of its own (widened until at least 30 donors), discarding draws that would violate the provision's upper bound. I repeat this 200 times using the approximate Bayesian bootstrap–each replication reweights the donor pool with uniform Dirichlet weights so the spread reflects estimation uncertainty in the calibration distribution. The corpus's median enactment year moves from 1987 (censored provisions at their lower bounds) to 1992.

The whiskers in Figure 1 mark the 2.5th–97.5th percentiles of each decade's count across the 200 imputations. They are narrow because the exactly dated mass never moves, each draw is confined to its own interval, and a decade count sums roughly 1,234 near-independent draws whose idiosyncratic randomness largely cancels. Of course, later-amended sections may have entered their containers on a different schedule than the never-amended ones used as a calibration sample. I validate this directly by recovering true enactment years for 27 of 50 sampled censored provisions (stratified by container era) from Westlaw's full legislative-history credits. The true lags track the calibration distribution—median 0 vs. 0, mean 12 vs. 11, 90th percentile 42 vs. 42—with somewhat less mass exactly at the container floor (52% vs. 63%).[2] True years sit slightly late relative to the draws, so any residual error leaves the imputed histogram marginally too old.

2 Who Participates

2.1 Sensitivity of Representativeness Estimates to Match Failure

Public commenter demographics rely on matching speaker names to the L2 California voter file. 67% of named non-government speakers match the file (107,496 of 161,008). If the unmatched 33%—non-registrants, out-of-town speakers, garbled or too-common names—differ demographically from matched speakers, the participant–voter gaps could be biased. This section bounds this possibility. Table 1 reports, for each headline variable, the matched mean and observed gap, worst-case (Manski) bounds, the breakdown point—the unmatched mean that would close the gap to exactly zero—and the gap recomputed with the unmatched mean imputed from unmatched speakers' own names (described below).

Variable	Matched mean (1)	Observed gap (matched − voters) (2)	Worst-case gap All = 0 (3)	All = 1 (4)	Erases gap if unm. mean = (5)	Name-implied gap (6)
White non-Hispanic	62.3	+16.4	-4.3	+28.9	13.0	+15.9
Female	47.8	-4.8	-20.7	+12.5	62.2	-4.9
Democrat	53.7	+5.5	-12.3	+20.9	37.2	–
Age (years)	52.6	+4.5	-3.0	+10.2	39.2	+4.2

Table 1: Sensitivity of participant–voter gaps to L2 match failure. Columns (1) and (5) are levels (shares in percent; age in years); gap columns (2)–(4) and (6) are in percentage points (age: years). Column (2) is the matched-participant mean minus the registered-voter mean. Columns (3)–(4) recompute the gap assuming all 33.2% unmatched speakers take the extreme value (0/1 for shares; mean 30/70 for age). Column (5) is the breakdown point: the mean the unmatched pool would need in order to close the gap to exactly zero—compare it with the matched mean in column (1). Column (6) recomputes the gap with the unmatched mean imputed from names—first-name →gender and first-name→birth-cohort lookups (see footnote).

If *zero* unmatched speakers were White non-Hispanic, the +16.4 pp European gap would flip to -4.3 pp. But the breakdown points show how implausibly extreme the unmatched pool would have to be: to erase the European gap, only 13% of unmatched speakers could be White non-Hispanic—versus 62% among matched speakers and 70% of unmatched speakers whose *names* are classified as European-origin; to erase the female gap, unmatched speakers would have to be 62% female (matched: 48%).

[2]The percentile of each true year within its own 200 imputations (its probability integral transform, mid-ranked across tied draws) should be uniform on [0, 1] if the imputation distribution is correct (Gneiting et al., 2007), and the observed mean of 0.57 does not reject uniformity ($z = 1.3$, $p = 0.20$). The central 95% band of the draws covers 24 of the 27 true years; two of the three misses lie above the band, one below. The right tail is heavier than uniform—5 of 27 true years (19%) sit above the 95th percentile of their own draws, against 5% expected—though three of the five remain inside the central band, whose upper edge is the 97.5th percentile.

The age gap is the most fragile: a mean unmatched age of 39 (matched: 53) would suffice. The name-based imputation in column (6) disciplines this directly.[3] Names are informative about gender, birth cohort, and European origin; the imputation yields an unmatched pool that is 47% female, 61% White non-Hispanic, with mean age 52. Because the surname lookup only covers surnames seen among matched speakers, an independent calibration covering all names—scaling the matched L2 share by the unmatched/matched ratio of ethnicolr predicted European-origin shares—gives a similar 59%. Under these estimates, every gap retains its sign and nearly all of its magnitude; in particular the age gap, fragile under the bounding arithmetic, is supported by the birth-cohort content of unmatched first names (+4.2 years implied versus +4.5 observed). I cannot impute political party, but I report the breakdown point (37%) for completeness.

I can also compare unmatched speakers by how they speak. Table 2 compares matched and unmatched named speakers on participation intensity, venue, comment content (three-theory classification), and land-use attitudes.

	Matched	**Unmatched**	**Difference**
Meetings spoken at (per name-city)	1.25	1.38	-0.14***
Comments per meeting	1.30	1.36	-0.05***
Words per comment	210	175	35***
Comment on agendized item	0.609	0.540	0.070***
Comment in unagendized public comment	0.183	0.142	0.040***
Preference expression (D)	0.684	0.526	0.158***
Accountability (A)	0.212	0.160	0.051***
Localized expertise (E)	0.815	0.746	0.069***
Land-use sentiment score	0.049	0.152	-0.103***

Table 2: Behavior of matched vs. unmatched named non-government speakers. Comment-level rows use the comment universe (141,258 matched / 119,000 unmatched comments). D/A/E are the three-theory classification detailed below (preference expression, accountability demand, localized information). Land-use rows use pro/con scores on land-use issues detailed below (9,285 / 5,773 comment-level scores). "Meetings spoken at" overstates repetition for unmatched speakers, since common names—which fail to match *because* they are ambiguous—also collapse distinct people into one name. Speakers with no usable name (1.3% of comments) are excluded. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. (Welch).

Unmatched speakers' comments are shorter, less likely to express a policy preference or convey localized information, and more pro-development on land-use items. This is the pattern one would expect if the unmatched pool mixes ordinary residents with non-registrants and out-of-town professionals—project applicants, developers' consultants—who cannot match a local voter file by construction and, to the extent they are non-resident, arguably fall outside the resident-participation estimand altogether.

[3] I use standard Bayesian name-based demographic imputation—surname→race as in BISG (Elliott et al., 2009) and its voter-file implementation (Imai and Khanna, 2016), first-name→sex and birth cohort as in Blevins and Mullen (2015)—except that I estimate the name dictionaries on our own matched sample (with empirical-Bayes shrinkage toward the matched mean, pseudo-count 10) so the mapping targets the L2 fields and the California population under study. The identifying assumption—that conditional on name, unmatched speakers resemble matched speakers—is the same. The independent European-share calibration uses the `ethnicolr` name-origin classifier (Sood and Laohaprapanon, 2018) used to estimate balance in Martin and Venugopal (2026).

2.2 Organizational Affiliations Among Public Speakers

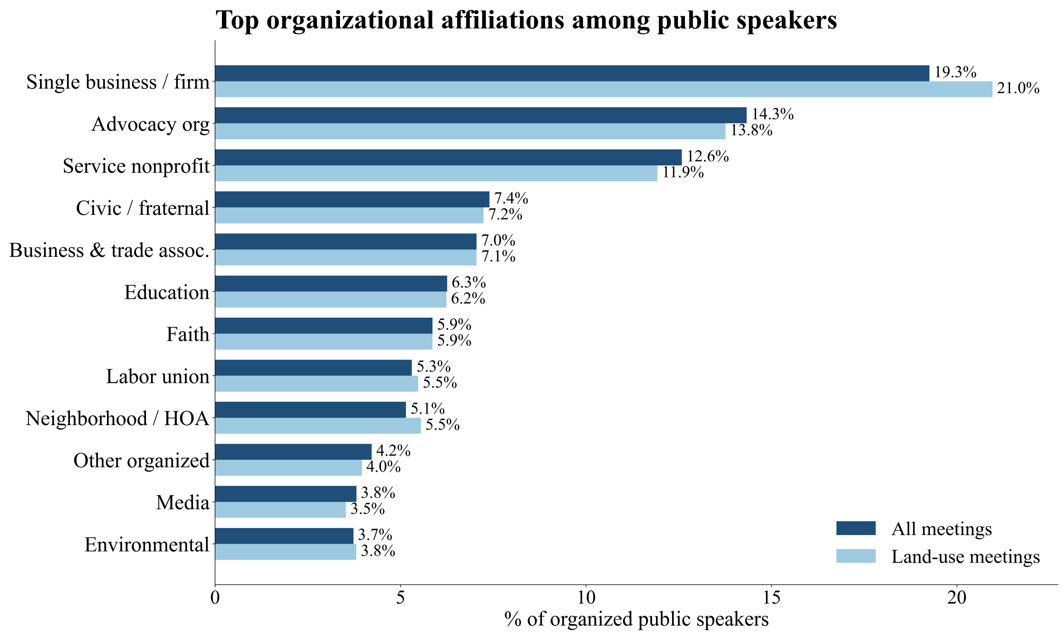

Figure 1: Organizational Affiliations Among Public Speakers

Organizational affiliation was collected at the speaker identification stage. For more details on this, see Martin and Venugopal (2026).

2.3 Municipal Turnout Demographics

	(1) Municipal-election voters	(2) 2021 off-year consolidated general
European origin	6.3**	9.2***
	(2.3)	(1.1)
Age 50+	−8.6***	−12.4***
	(2.3)	(1.1)
Democrat	0.6	2.1
	(2.3)	(1.1)
Female	−6.5**	−3.5**
	(2.4)	(1.1)
Homeowner	−4.8*	−7.4***
	(1.9)	(1.0)
Participants (N)	454	2,007
Local voters (N)	17,424	83,604
Cities	4	8
City fixed effects	Yes	Yes

Table 3: Meeting Participant vs. Local Voter Demographics: Within-City Differences in Means

Notes: Each cell is the within-city difference in means (in percentage points) between meeting participants and non-participant local voters, estimated by OLS of the demographic indicator on a participant dummy and city fixed effects; positive values indicate participants exceed local voters. Heteroskedasticity-robust (HC1) standard errors in parentheses. I do not cluster by city: with only four (eight) cities, cluster-robust standard errors are unreliable, and identification comes from differences among tens of thousands of individuals within each city's election. Column (1) covers the four sample cities with standalone, off-cycle municipal-election coverage—San Leandro (November 2022) and Downey, Inglewood, and Vacaville (2023). Column (2) covers voters in the 2021 odd-year consolidated general election: Riverside, Santa Barbara, Inglewood, Carson, San Jose, Los Gatos, Hemet, and Rosemead. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

2.4 Benchmarking Against Metro-Boston Zoning Meetings

The canonical benchmark for participant representativeness is Einstein et al. (2020). Table 4 places their headline results beside the equivalent quantities in my sample.

	This sample (California)			Einstein et al. (Boston)		
Variable	Commenters	Voters	Gap	Commenters	Voters	Gap
Female	47.8	52.6	-4.9	43.3	51.3	-8.0
Democrat	53.7	48.2	+5.5	32.0	31.7	+0.2
White	62.3	45.9	+16.4	95.0	86.7	+8.2
Over 50	57.9	45.7	+12.2	75.0	52.6	+22.4
Homeowner	79.0	74.7	+4.3	73.4	45.6	+27.8

Table 4: Commenter and registered-voter demographics: California city council meetings vs. metro-Boston planning and zoning meetings. Boston columns reproduce Table 1 of Einstein et al. (2020); California columns are computed from the L2 voter file as in the main text. Shares in percent; gaps in percentage points. "White" is White non-Hispanic (L2 European origin) in my sample and voter-file race in theirs. Homeownership definitions differ across the two studies and levels are not comparable; see text.

The two samples differ particularly on race and politics, with voters in California being much more Democrat and much less White. On an 87%-White electorate the largest possible White gap is 13 points; a racial gap of my size (+16.4) could not arise in their sample. The homeowner comparison is bounded in the opposite direction: conditional on a property-record match, 74.7% of my registered voters are homeowners, capping the feasible gap at 25.3 points—below their observed +27.8.[4]

Table 5 recomputes the five gaps under progressively tighter restrictions toward their setting: land-use commenters (housing and land-use topics), the 15 demographically Boston-like cities in my sample, and project-specific (adjudicative) land-use items. The Boston-like electorate resembles theirs on age and gender (55.0% over 50 vs. their 52.6%) and is far closer on race (70.8% White non-Hispanic).

	Full sample	**Boston-like cities**	**Land use, all cities**	**Land use, Boston-like**	**Adjudicative, Boston-like**	**Einstein et al. (Boston)**
Female	-4.9	-4.7	-8.3	-8.3	-8.7	-8.0
Democrat	+5.5	+7.8	+4.7	+7.7	+8.6	+0.2
White	+16.4	+12.0	+20.2	+14.7	+13.6	+8.2
Over 50	+12.2	+15.0	+14.6	+15.6	+14.1	+22.4
Homeowner	+4.3	+5.0	+3.8	+3.7	+3.5	+27.8

Table 5: Commenter–voter gaps under restrictions toward the Einstein et al. (2020) setting

Notes: Each cell is the pooled difference in means (percentage points) between commenters and non-participant registered voters in the indicated universe. Land-use commenters are named, L2-matched non-government speakers on housing and land-use voting issues (distinct people, earliest appearance); adjudicative items are project-specific decisions (variances, conditional-use permits, site-specific approvals) as distinct from general-applicability measures, per an LLM classification of land-use items. Boston-like cities are the 15 of 115 with population under 100,000, owner-occupancy at or above 55%, and majority-White populations (ACS place means): Citrus Heights, Encinitas, Folsom, Laguna Niguel, Lake Forest, Livermore, Los Gatos, Menlo Park, Mission Viejo, Napa, Novato, Palo Alto, San Clemente, Vacaville, Walnut Creek. Commenter Ns: 107,788 person-years (full), 11,196 (Boston-like); 13,687 people (land use, all cities), 1,477 (land use, Boston-like), 925 (adjudicative, Boston-like). Maximum binomial standard error in the smallest column is roughly 1.6 points.

[4] Homeownership also differs in *measurement*. My indicator matches a voter's address to CoreLogic parcel records (situs or owner mailing address) and codes the voter a homeowner if any matched parcel is owner-occupied: it is a household-level tenure measure that counts non-owner adults living in owner-occupied homes, and it is defined only for the 68% of voter person-years (80% for commenters) that match a parcel at all. Because apartment and post-office-box addresses match at lower rates, the matched baseline overstates ownership and the match-conditional gap is likely an understatement: assuming every unmatched person rents widens the gap from +4.3 to +11.9, while assuming every unmatched person owns shrinks it to +0.6. In relative terms, homeowners in my sample are 1.3 times as likely to comment as non-homeowners, against "more than three times" in theirs, which relies on person-level matches to assessor records. Both the measurement gap and the venue difference below plausibly contribute.

3 What Participants Say

3.1 Annotation of Commenter Attitudes Toward Development

Prompt

```
You are given: -Issue title: {issue} -Issue summary: {summary}. Below is a dictionary where each key is a
     unique identifier for a comment on THIS Issue, and each value is the transcribed speech text for that
     comment. Your task is to analyze each value (comment text) and predict whether the speaker is in favor of
     or opposed to a more development/growth outcome on this issue on a continuous scale of 1 to -1 where: 1.0
     = clearly supportive of more development/growth (e.g., favors approval of a project, wants more
     development, etc.), 0.0 = neutral / unclear, -1.0 = clearly opposed to more development/growth (e.g.,
     wants more restrictions on new development, has a lot of concerns about development). Return **only** a
     list of lists, where each inner list has exactly two elements: [unique_id, score]. Unique_id must be a
     string [-1, 1], score must be a float in [-1, 1].
```

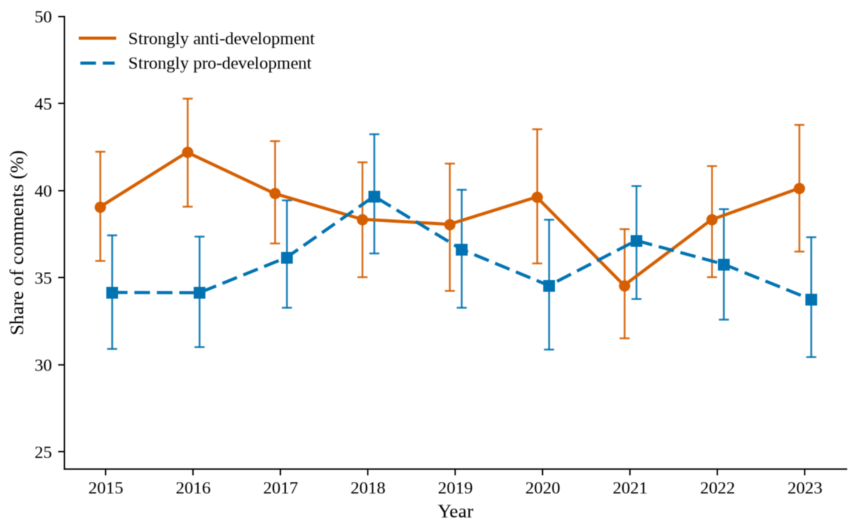

Figure 2: **Development attitudes of ordinary public commenters, 2015–2023.** Yearly shares of public comments scored clearly opposed to (≤ -0.5, solid) or clearly supportive of ($\geq +0.5$, dashed) a more development-oriented outcome, on the model's -1 to $+1$ stance scale. The sample is the 26,573 development-attitude comments from Appendix 3.1 dated 2015–2023: comments on land-use and housing items with a clear build-more-versus-less axis, by non-government speakers, excluding applicant-, developer-, and business-affiliated commenters and rent-control/tenant-protection items. Error bars are 95% percentile intervals from a within-year block bootstrap (2,000 replications) that resamples *hearings* rather than comments, reflecting the clustering of comments within contentious items. Because the roughly seventy design-weighted human labels cannot support year-level debiasing, the plotted series are the model's aggregates; the corpus-level correction in Table 7 shifts the *level* of both series down by about three percentage points with the uncertainty reported there, while year-to-year comparisons assume the model's measurement error is stable over time. 2024 is omitted as a partial year.

Validation. I validate the stance scores against blind hand labels from two coders. I labeled 100 land-use and housing comments and a research assistant labeled 121, both of us blind to the model's scores, with comments drawn to span the model's score range, topics, and procedural categories (appeals, historic-preservation matters, housing-element items). Fifty comments were labeled by both of us with the overlap hidden.[5] After dropping "unclear" responses and a handful of frame errors caught in review (dais speech misattributed to the public, rent-board matters), 99 of my labels, 100 of the RA's, and 49 of the overlap pairs enter the comparisons.

Agreement is summarized in Table 6. The production model (Gemini 3 Flash) matches my three-way pro/neutral/opposed collapse on 73% of comments ($\kappa = 0.58$), with Pearson $r = 0.80$ on the -1 to $+1$ scale, sign agreement of 80%, and a mean

[5]These were drawn by stratified random sampling from the comment universe, with strata crossing five item domains (housing-element items; legislative and adjudicative items in the land-use and housing topics) with the model's predicted pro/neutral/opposed class, oversampling rare cells, so each label carries a known sampling weight.

absolute difference of 0.32. On the 49 blind-overlap comments, the RA and I agree with each other 80% of the time ($\kappa = 0.68$, $r = 0.83$)—so the model's disagreement rate with a human coder is close to the rate at which two trained human coders disagree. Relative to my labels the RA reads systematically more pro-development, while the model tends to misread angry *supporters* ("you're just diddling around, approve it") as opponents.[6]

Table 6: Agreement on commenter development stance

	n	Three-way / κ	Pearson r	Sign agr.	MAD
Model vs. author	99	73% / 0.58	0.80	80%	0.32
Model vs. RA	100	70% / 0.54	0.70	69%	0.41
Author vs. RA (blind overlap)	49	80% / 0.68	0.83	80%	0.23

Notes: Three-way collapses the $[-1, 1]$ score at ± 0.5 into pro / neutral-or-mixed / opposed; MAD is the mean absolute difference on the continuous scale.

Uncertainty. Averaging the model's scores over the corpus estimates the model's mean, not the value a human coder would find. I therefore report prediction-powered inference (PPI) estimates (Angelopoulos et al., 2023a,b) of the corpus-level attitude aggregates, which debias the model mean using the human-labeled subset $\mathcal{L}$:

$$\hat{\theta} \;=\; \bar{Y}_{\mathcal{L}} \;+\; \lambda\big(\bar{f}_{\mathcal{U}} - \bar{f}_{\mathcal{L}}\big), \tag{1}$$

where Y_i is the human label for unit i, f_i the model's label, $\mathcal{U}$ the full corpus, and the power-tuning parameter λ the slope from regressing Y on f within $\mathcal{L}$. Read $\hat{\theta}$ as the human-labeled mean transported to the corpus by the model: the parenthetical term is the model-measured gap between the corpus and the labeled sample, and λ scales that correction by how informative the model is about the human label. A model whose labels are pure noise yields $\lambda = 0$, collapsing $\hat{\theta}$ to the labeled-sample mean—so a weak model costs precision, not validity. Here $\mathcal{L}$ is the 74 of my labels that fall in the stratified design sample, entering with inverse-stratum response weights; Y and f are the continuous scores for the mean-stance estimand and clearly-supportive or clearly-opposed indicators for the share estimands; and confidence intervals come from a stratified bootstrap. Table 7 reports the results. For all three estimands—mean stance, the share of comments clearly supportive, and the share clearly opposed—the naive model aggregate lies inside the debiased 95% confidence interval, and the anti-minus-pro gap is essentially unchanged. I compute the anti minus pro gap's confidence interval within each bootstrap replicate, which accounts for the correlation automatically. The resulting 95% interval, $[-14.1, +17.8]$ percentage points, straddles zero (35% of replicates place the gap at or below zero), so the debiased corpus is statistically indistinguishable from an even split between clear supporters and clear opponents. Individual scores remain noisy, so I continue to interpret only aggregates.

Table 7: Prediction-powered estimates of corpus-level attitude aggregates

	Naive	PPI	95% CI	λ
Mean stance (-1 to 1)	−0.018	+0.045	[−0.083, 0.174]	0.61
Pro-development share	35.9%	33.0%	[25.6, 41.9]	0.71
Anti-development share	38.8%	35.6%	[25.5, 46.1]	0.49
Anti − pro gap (pp)	+2.9	+2.7	[−14.1, +17.8]	—

Notes: The universe is the 27,165 public land-use and housing comments used in the attitude-composition analysis: comments by non-government speakers, with applicant-, developer-, and business-affiliated commenters excluded (via a high-recall—though not perfect—applicant regex over the comment text plus the business-affiliation code from the speaker-identification stage), after dropping 652 rent-control and tenant-protection comments (of 27,817) whose stance does not map onto a development attitude. "Naive" averages the production model's score over that universe; "PPI" debiases it with the author's 74 design-weighted labels (power-tuned λ; Angelopoulos et al., 2023a,b). CIs are percentile intervals from a stratified bootstrap (2,000 replications) over the sampling strata. The gap row differences the anti and pro shares within the same bootstrap replicate, so its interval reflects their correlation. My labels drawn outside the stratified design sample enter the agreement statistics but not the weighted PPI estimates.

3.2 Annotation of Comment Characteristics

The three purposes annotated here correspond to the three (non-intrinsic) theories of participation developed in the main text. The partition is also data-driven. Before fixing the taxonomy, I ran LLM-based unsupervised clustering on a random sample

[6] My labels were entered blind, but after an interim scoring pass I reviewed the largest model–author disagreements with a higher-reasoning model (Claude Fable 5) and revised five of my labels; as in the project-conditions validation, I sided against the model on several rows and with it on others, so the final labels are not simply anchored to either annotator. Four of the 49 overlap rows had been discussed in that review before my labels were finalized.

of 500 land-use comments without supplying ex ante categories. Six communicative intents emerged: urging approval or rejection, proposing modifications or conditions, requesting information or transparency, sharing lived experience, demanding accountability, and advocating broader policy direction. These inductively discovered categories map relatively well to the theories; advocacy intents map to preference expression, demands for accountability and transparency onto accountability, and sharing of lived experience onto expertise. The prompt below operationalizes the three purposes as non-mutually-exclusive binary labels.[7]

Prompt

```
SYSTEM PROMPT: You classify communicative purposes of public comments from California city council meetings.
     For each comment, determine whether EACH of three purposes is present (1) or absent (0). A comment can
     serve multiple purposes simultaneously.
(D) Preference Expression: The commenter is expressing a preference about what the council should do  urging
     approval or rejection, stating support or opposition, advocating for a particular outcome. The core intent
      is: "I want you to do X" or "I support/oppose Y." This includes comments that give reasons for their
     preference, as long as the primary thrust is telling the council what outcome the speaker wants.
(A) Accountability: The commenter is holding officials, staff, or developers accountable  questioning the
     fairness of a process, alleging broken promises or procedural violations, demanding explanations or
     transparency, pressing officials to justify their actions, or flagging that something has gone wrong that
     higher authorities should know about. The core intent is: "You need to answer for this" or "This process
     is not being conducted properly." Examples of A=1: alleging a violation of state housing law, citing a
     broken campaign promise, demanding a formal review or investigation, flagging that a required procedural
     step was skipped. Examples of A=0: expressing strong opposition, criticizing a decision as bad policy,
     pointing out that a staff report fact is wrong, making pointed or hostile comments without explicitly
     invoking accountability, failure of duty, or procedural violation. Note: neutral requests for
     clarification that do not imply wrongdoing, broken promises, or procedural failure should NOT be coded A=1
       a speaker simply asking how a process works or seeking information is not holding anyone accountable.
(E) Localized Expertise: The commenter is providing specific factual information  citing data, describing
     conditions, reporting observations, or offering technical or experiential knowledge. The core question is:
      does the comment contain identifiable factual claims beyond pure opinion or preference? If yes, E = 1. Do
      not filter based on whether the council already knows the information; that assessment happens at the
     claim level.
Examples for E (Localized Expertise):
Clear E:
IS E: "11% of Pleasanton's population was born in Commonwealth countries."  Specific demographic statistic.
IS E: "I spend three hours commuting for my son to practice cricket."  Firsthand quantified experience.
IS E: "CIF has made beach volleyball a high school sport."  Specific external institutional fact.
IS E: "Unintentional child firearm deaths rose 43% during COVID."  Named statistic from research.
IS E: "Thomas Fallon was not the first person to raise the U.S. flag over San Jose."  Specific historical
     correction.
Borderline  still E:
IS E: "The ball stops in the thick grass on baseball pitches, so kids can't play ground shots."  Sounds like
     opinion but is a specific observable claim about field conditions affecting gameplay.
IS E: "There are 20 porta potties and 10 washing stations at Phase 3 with no lighting."  Counts of city-
     provided amenities  the city should know this, but the ground-truth operational report from a resident
     still qualifies as a factual claim.
IS E: "We've seen a lot of 404 errors and broken links on the city website."  The city could test its own site,
      but the resident is reporting a specific user experience.
Clear NOT E:
NOT E: "Cricket is a great sport and now is the time to invest."  Evaluative assertion, no factual claim.
NOT E: "I love the city and have lived here four years."  Background context, not a factual claim about
     conditions.
NOT E: "Please prioritize the cricket field at the top for this year."  Pure preference/request.
NOT E: "We need to do the right thing and declare a climate emergency."  Normative statement without factual
     content.
NOT E: "This city should be ashamed of itself."  Evaluative, no specific factual claim.
When E = 1, extract up to 5 factual claims from the comment. Selection rule: Prioritize claims that are most
     specific and verifiable  prefer a named statistic, date, location, or observable condition over a general
     assertion. If fewer than 5 factual claims are present, return only those found. Use the decision_context
     provided with each comment to assess decision relevance. If no decision_context is provided, assess
     relevance based on the issue label. If the comment is labeled "(General public comment  no specific agenda
      item identified)", set decision_relevant = 0 for all claims, since there is no identifiable decision to
     be relevant to.
```

[7] To bound output costs, comment text passed to the classifier is truncated to its first 1,200 characters, which binds for 38 percent of comments (median comment length is 975 characters).

```
For each claim, provide: "claim": A short paraphrase of the factual claim (not verbatim transcript text); "
     firsthand": 1 if the speaker is reporting something they personally witnessed, experienced, or directly
     measured  e.g., "I saw police cars there at 9pm," "I found syringes in the gutter," "I've called the
     police five times." 0 if the claim is based on documents, what others said, general local knowledge, or
     facts about a place the speaker knows without having personally observed the specific condition being
     claimed; "expert_basis": 1 if the speaker explicitly invokes professional credentials, organizational role
     , or technical knowledge as the basis for the claim, 0 otherwise; "external_data": 1 if the claim
     references a specific statistic, study, dataset, or named source external to the city  0 otherwise. Citing
      the city's own staff reports or agenda materials back to the council does not count; "decision_relevant":
      1 if the claim bears directly on the specific decision described in the decision_context or issue label
     there must be a clear, concrete link between the claim and what the council is deciding, not merely that
     the claim *could* be useful background. 0 otherwise. When in doubt, code 0; "corrects_record": 1 if the
     claim explicitly contradicts, corrects, or challenges something staff or officials have stated, 0
     otherwise; "site_specific": 1 if the claim describes conditions at a named geographic location or facility
     , 0 otherwise; "verifiable": 1 if a neutral third party could in principle confirm or refute the claim
     using records, measurement, or direct observation  the claim must be specific enough to be falsifiable. 0
     if the claim is vague, subjective, or unfalsifiable even in principle. IS verifiable: "Crime rose 43% last
      year" (checkable against stats), "There are 20 porta potties at Phase 3" (countable on site),"The
     commission voted unanimously" (checkable against minutes). NOT verifiable: "The meeting was chaotic" (
     subjective threshold), "The project will hurt the neighborhood" (speculative prediction), "There was a
     lack of respect for elders" (subjective judgment); "asymmetric": 1 if the specific claim would NOT
     typically appear in a staff report, agenda memo, environmental review, or other materials the council
     receives  0 if the council's standard information sources would already include this. Ask: would a well-
     prepared council member already know this from their briefing materials? If yes, code 0. Note: a claim can
      be firsthand or from an external source and still be non-asymmetric (e.g., a geographic fact that appears
      in the site plan); "asymmetry_reasoning": One short phrase (100 chars) explaining why the council would
     or would not already know this from their standard briefing materials.

Each comment will be provided with an issue label and, where available, a decision_context describing what the
     council is specifically deciding. Reply with ONLY a JSON object

(Each API call sends up to 10 comments. The user message opens with the line below, then repeats the per-
     comment block for each comment.)

PER-COMMENT PROMPT: Below are the comments. For each, determine whether each of the three purposes (D, A, E) is
      present (1) or absent (0).
--- Comment id=<comment_id> ---
[<idx>/<total>] <SPEAKER_ID> | <N> words
Issue: <issue title>
Decision context: <issue summary, truncated to 400 chars>
<comment text, truncated to 1200 chars>
(When no agendized+public issue is attached, the "Issue:" line instead reads: Issue: (General public comment
     no specific agenda item identified) and no "Decision context:" line is shown.)
```

Validation. I validate the LLM's annotation of comments with the three non-mutually-exclusive broad purposes (preference expression, accountability, and expertise sharing), and factual claim attributes against hand labels on a sample of 135 public comments drawn from 15 randomly sampled council meetings.[8] I coded each comment blind to the model's output. I also compare my labels to those of a research assistant and two more advanced models, GPT-5.5 and Claude Opus 4.7.

The production model (Gemini 3 Flash) matches my labels 87% of the time on preference expression ($\kappa = 0.68$), 90% on accountability ($\kappa = 0.53$), and 93% on expertise ($\kappa = 0.67$). The frontier panel does not systematically outperform Gemini against my labels, and models disagree with each other about as often as they disagree with me. The most common disagreements are borderline cases in the expertise context (12 of 40), which include incidental factual content inside appreciation comments (*"residents that speak Zapotec, Mixtec, and Triqui"*) that the models tend to count and I did not, and, conversely, experiential institutional knowledge that only some readers count as factual.

Uncertainty. Averaging Gemini's labels over all 263K classified comments estimates Gemini's mean, not the share a human coder would find. I therefore report PPI estimates of the corpus-level purpose shares, applying the estimator of equation (1) with the 135 human labels as $\mathcal{L}$. Both Y and f are binary here, so the power-tuning parameter λ reduces to $\Pr(Y = 1|f = 1) - \Pr(Y = 1|f = 0)$: how much more likely a human positive is when the model says 1 than when it says 0. Table 9 reports the results. The naive model share of preference comments (61.2%) understates the human-protocol share

[8] The underlying sample is 50 meetings drawn at random from the transcript corpus with a cap of two meetings per city–year, so that high-frequency councils (e.g., Los Angeles) do not dominate the sample; the 135 comments coded to date, labeled in a random order over all public comments in those meetings, span 15 of them.

(69.6%, 95% CI [64.3, 74.9]), while the naive accountability share (18.9%) overstates it (15.9%, CI [11.1, 20.7]). Because the validation comments cluster within 15 meetings, I also report a meeting-level bootstrap interval. Finally, the conclusion does not appear to depend on which LLM annotator we used as the estimates acrosss Gemini and other LLMs are relatively similar.

I apply the same two checks—agreement and prediction-powered uncertainty—to the seven binary attributes the model attaches to each extracted factual claim. The unit here is the claim rather than the comment: 151 claims blind-labeled across 62 comments (Table 10). Agreement ranges from $\kappa = 0.44$ (expert basis) to $\kappa = 0.79$ (firsthand); corrects-record positives are rare (4.0% of claims), so its κ is imprecise and should be read alongside the raw counts in the table. Table 11 reports the same PPI estimator at the claim level.

Table 8: Agreement on the three-purpose classification (percent agreement / Cohen's κ)

	Preference (D)	Accountability (A)	Expertise (E)	n
Human positive share	76%	13%	87%	135
vs. human (author) labels				
Gemini 3 Flash (production)	87% / 0.68	90% / 0.53	93% / 0.67	135
GPT-5.5	90% / 0.75	91% / 0.57	92% / 0.52	135
Claude Opus 4.7	90% / 0.74	91% / 0.57	86% / 0.53	135
Second human coder (RA)	93% / 0.83	95% / 0.31	65% / 0.20	85
between models (full sample)				
Gemini–GPT-5.5	88% / 0.73	95% / 0.78	94% / 0.69	465
Gemini–Opus	88% / 0.75	97% / 0.83	83% / 0.51	460
GPT-5.5–Opus	91% / 0.82	97% / 0.86	80% / 0.40	460

Notes: Human rows compare against the author's blind labels on 135 comments; the second human coder covers the 85-comment subset.

Table 9: Prediction-powered estimates of corpus-level purpose shares

	Naive	PPI	95% CI	Cluster CI	Model range
Preference (D)	61.2%	69.6%	[64.3, 74.9]	[64.0, 88.6]	[68.1, 71.9]
Accountability (A)	18.9%	15.9%	[11.1, 20.7]	[10.8, 36.3]	[12.0, 13.1]
Expertise (E)	78.4%	77.2%	[73.0, 81.4]	[67.9, 79.3]	[82.6, 83.6]

Notes: "Naive" is the uncorrected Gemini mean over all 263,641 classified comments. "PPI" debiases it with the 135 human labels (power-tuned λ; Angelopoulos et al., 2023a,b). "Cluster CI" is a percentile bootstrap resampling the 15 labeled meetings. "Model range" spans the debiased estimate of the validation-sample mean when GPT-5.5 or Claude Opus 4.7 replaces Gemini as the surrogate labeler.

Table 10: Agreement on extracted-claim attributes

Attribute	Human +	Gemini +	Agreement	κ	Disagreements
Firsthand	38.4%	47.7%	89.4%	0.79	16
Expert basis	13.9%	16.6%	85.4%	0.44	22
External data	10.6%	17.9%	87.4%	0.49	19
Decision relevant	49.0%	53.0%	88.1%	0.76	18
Corrects record	1.3%	4.0%	94.7%	-0.02	8
Site specific	33.1%	47.7%	85.4%	0.70	22
Asymmetric	69.5%	69.5%	84.1%	0.62	24

Table 11: Prediction-powered estimates of corpus-level claim-attribute shares

	Naive	PPI	95% CI	Cluster CI	n_{eff}	Model range
Firsthand	43.1%	34.8%	[30.1, 39.5]	[23.7, 38.4]	419	[29.3, 37.6]
Expert basis	25.7%	17.7%	[12.7, 22.6]	[4.6, 31.9]	187	[11.4, 14.8]
External data	13.5%	8.8%	[4.6, 13.0]	[4.8, 18.1]	205	[8.1, 14.2]
Decision relevant	44.2%	42.3%	[37.1, 47.4]	[37.4, 54.6]	363	[46.2, 55.7]
Corrects record	3.3%	3.5%	[1.2, 5.9]	[0.0, 7.8]	263	[3.0, 8.3]
Site specific	40.0%	27.7%	[22.7, 32.8]	[18.8, 31.1]	331	[29.0, 34.5]
Asymmetric	58.8%	62.8%	[57.1, 68.6]	[57.1, 71.9]	248	[65.0, 71.8]

Notes: Same estimator as Table 9, at the level of the extracted factual claim. "Naive" is the uncorrected Gemini share over all 459,457 claims extracted from the classified corpus; "PPI" debiases it with the 151 author-labeled claims (power-tuned λ). "Cluster CI" is a percentile bootstrap resampling the 15 meetings the labeled claims come from. "Model range" spans the debiased estimate of the validation-sample mean when GPT-5.5 or Claude Opus 4.7 replaces Gemini as the surrogate labeler. Because each model extracts its own claims, the author's labels (given on Gemini's claims) are transferred to panel claims by within-comment text matching (best match, similarity $\geq$ 0.6); GPT-5.5 matches 115 and Opus 80 of the 151 labeled claims.

3.3 Asymmetry-Specific Audit

The audit validates the per-claim asymmetric flag against briefing materials council members likely received. The sampling frame is land-use voting issues from 2022 onward that were agendized, decided by a non-unanimous roll call, and project-specific. Each meeting whose briefing documents could be retrieved from the city's public legislative archive was audited: 28 meetings in 13 jurisdictions, covering 449 public-comment claims from those meetings that the classifier had marked decision-relevant. For each meeting, the full agenda packet and the item's attachments—staff reports, exhibits, CEQA/EIR documents, and attached studies—were downloaded from the jurisdiction's public legislative archive (the Legistar web API for most cities). Attachments named as public-comment correspondence were excluded. Each claim, together with the complete document set as native PDFs, was then given to a separate LLM auditor (Gemini 3 Flash, temperature 0), which scored the claim 0 (not in the packet and not reasonably inferable), 1 (inferable from the packet or expected general knowledge), or 2 (clearly stated in the packet), quoting the packet passage as evidence for any score of 2. Document sets over 600 pages were pruned by dropping appendix-style files first.

Of the 186 claims tagged asymmetric, 133 (71.5%) were confirmed absent from the packet and only 34 (18.3%) were in fact clearly stated in it. In the other direction, 75 (28.5%) of claims tagged not asymmetric were actually absent from the packet, i.e., the classifier more often credits the council with knowledge the briefing materials do not contain than the reverse, implying the corpus-wide asymmetric share is, if anything, an underestimate. Across all 449 audited claims, 46% were absent, 16% inferable only, and 38% clearly stated.

3.3.1 Accountability Register Classification

Every comment flagged A= 1 on an agendized land-use item is classified against the seven non-mutually-exclusive accountability registers described in the main text, which were themselves induced from the data as described in the main text. The classification prompt is below.

Prompt

```
SYSTEM PROMPT: You classify the ACCOUNTABILITY REGISTER(S) of public comments from California city council LAND
     -USE meetings. Every comment here already contains accountability content (holding officials, staff, or
     developers to account). Your job is to identify WHICH of 7 registers each comment uses. A comment may use
     several  label each 1 (present) or 0 (absent). (1) procedural_fidelity  Demands that formal RULES and
     legal PROCESS be followed: adequate notice, proper hearings, CEQA as a required procedural step, Brown Act
     , validity of urgency/emergency declarations, due process. The move is "the correct procedure was not
     followed." (2) epistemic_misrepresentation  Alleges officials/staff/developers MISREPRESENTED, manipulated
     , cherry-picked, hid, fabricated, or used knowingly false/bad-faith data, studies, or analysis to justify
     a decision. The move is "they are misleading you / the evidence is dishonest"  there must be a SPECIFIC
     imputation that the evidence is false or used in bad faith. The following are NOT this register: (a) a
     good-faith factual disagreement ("I think the traffic number is too low"), OR a claim that an analysis is
     merely INADEQUATE / INCOMPLETE / legally insufficient under CEQA (failed to study an impact, wrong
     document type, must be recirculated, no study was done)  that is procedural_fidelity unless the comment
     alleges the underlying data is false or manipulated; (b) a bare pejorative label for an analysis ("the EIR
      is nonsense / boilerplate / a sham") with no specific falsity or concealment claim; (c) error or
     misinformation attributed to the SPEAKER themselves, to opponents, or to the public rather than to
```

```
    officials/staff/developers. (3) participatory_legitimacy  Alleges public engagement was bypassed,
    performative, predetermined, rushed, or restricted: "you've already decided," inadequate outreach, comment
    -time limits, ignoring community input. The move is "the public was not genuinely heard." (4)
    distributive_equity  Holds officials accountable for DISPARATE IMPACTS on marginalized or protected groups
    : racial bias, displacement, exclusionary zoning, environmental justice, double standards by neighborhood
    or class. The move is "this decision unjustly harms a vulnerable group." (5) performance_enforcement
    Demands the city or developer HONOR PRIOR PROMISES / conditions of approval / community-benefit agreements
    , or ENFORCE existing codes against ongoing violations. The move is "you committed to X / the law already
    requires X  follow through." (6) ethical_integrity  Alleges IMPROPER INFLUENCE or self-dealing: conflicts
    of interest, campaign-money/pay-to-play, developer favoritism, corruption, financial self-interest. The
    move is "this decision is corrupt / conflicted." (7) jurisdictional_authority  Contests WHO has authority
    to decide: state mandate vs. local control, preemption, administrative/staff overreach vs. elected or
    voter authority. The move is "this body lacks (or is ceding) the proper authority." Also pick the single
    PRIMARY register (the dominant move) and give a <=20-word rationale.
(Each API call sends up to 10 comments. The user message is the line COMMENTS: followed by a JSON array of {"id
    ": <comment_id>, "text": <comment text>} objects; comment text is passed in full, untruncated.)
```

Validation. I hand-labeled a blind random sample of classified comments plus an oversample of the two rarest registers (jurisdictional authority and ethical integrity). Seventy-six comments were fully labeled (63 random, 13 oversample). In the blind first pass, agreement was 79 percent (pooled $\kappa = 0.45$), with the two evidence-adjacent registers weakest (procedural fidelity and epistemic misrepresentation, both $\kappa = 0.24$). I then re-adjudicated every procedural-fidelity and epistemic-misrepresentation disagreement (39 comments) against the codebook with the full comment text and the model's rationale in view, revising my own label wherever the first-pass call did not survive a strict reading of the register definitions (23 of 46 disagreed register labels were revised; the remainder were upheld against the model). The other five registers were not re-adjudicated and retain their blind first-pass labels.

Table 12 reports post-adjudication results: pooled agreement is 83 percent ($\kappa = 0.53$; random core only, 85 percent, $\kappa = 0.58$). Procedural fidelity rises to $\kappa = 0.68$, and its residual disagreements are almost entirely one-sided: the model marks the register on 11 comments I do not, upheld cases in which it stretches "the correct procedure was not followed" to cover delay complaints, unfunded-mandate objections, and critiques of deliberative adequacy with no identifiable procedural rule at stake. The PPI-debiased procedural-fidelity share (equation (1)) is 33 percent against the model's 44, so the model-based share should be read as an upper bound.

Table 12: Blind validation of the seven-register classifier

	Agreement (n=76)			Corpus share (%)		
Register	Percent	κ blind	κ final	Model	PPI	95% CI
Procedural fidelity	84	0.24	0.68	44.3	33.2	[24.1, 42.3]
Epistemic misrepresentation	86	0.24	0.44	19.4	16.3	[7.3, 25.2]
Participatory legitimacy	84	0.66	0.66	36.2	33.0	[23.0, 43.0]
Distributive equity	78	0.41	0.41	29.5	42.2	[31.7, 52.7]
Performance enforcement	78	0.34	0.34	23.6	30.0	[18.3, 41.7]
Ethical integrity	80	0.39	0.39	14.9	18.1	[7.2, 28.9]
Jurisdictional authority	91	0.49	0.49	6.6	5.0	[0.0, 10.5]

Notes: Hand labels vs. Gemini 3 Flash on 76 comments (63-comment random core plus 13-comment oversample of the two rarest registers; agreement and κ use all 76, PPI shares use the random core only). "κ blind" is the blind first pass; "κ final," the agreement percentage, and the PPI columns use final labels after re-adjudication of all procedural-fidelity and epistemic-misrepresentation disagreements. CI is the 95 percent PPI interval.

4 Councilmember Citation Pipeline

A "citation" is a moment when a councilmember, while discussing a housing or land-use item, explicitly refers to something a member of the public said at that same meeting—for example, "As Ms. Alvarez mentioned, the driveway backs onto the school crossing," or "several residents tonight raised the noise issue." The measure counts the act of referring to a public comment, whether the member goes on to agree with it, act on it, or argue against it. I count citation occurrences through a four-step pipeline; Table 13 summarizes the funnel and the validation audit attached to each step, and the paragraphs below give details.

Table 13: Councilmember citation pipeline: funnel and validation summary

Step	Filter	Turns remaining	Validation (result)
—	universe: council turns on deliberated, commented housing/land-use items	3,810,386	—
1	at least ten words	2,179,219	audit (i), discarded short turns: ~390 missed citations, 4.5% of counted volume
2	matches one of seven cue families	25,844	audit (ii), never-flagged turns: ~2,330 missed citations, 27% of counted volume
3	light screen, then full-context model reading: relays another person × public speaker × this meeting	8,672	precision audit (iii): 77.1% of counted turns confirmed by human; false-omission probe (iv): 6.1% of rejected turns are true citations
4	(no filtering; links speaker, classifies purpose)	8,672	purposes audited below (purpose validation); speaker links audited with the analyses that use them

Notes: "Turns" are distinct diarized segments; counts pool the housing and land-use topic classes. The Step-2 row reports distinct flagged turns; the same turns appear in roughly 31,000 turn–item pairs, and fall within 20,852 of the 275,708 merged speaking turns (same speaker, gaps of at most one minute), leaving the 254,856 never-flagged merged turns audited in audit (ii). The Step-3 light screen and full reading use the same model; 14,307 of the 25,844 flagged turns reached the full reading.

Step 1 (universe and ten-word floor): Within city council meetings, I take housing and land-use agenda items on which the council deliberated and the public commented, and collect every councilmember speaking turn of at least ten words. The full set of councilmember turns on these items numbers 3.8 million; 57 percent (2.2 million turns) meet the ten-word threshold and enter the pipeline.[9]

Step 2 (cue-pattern filter): I conduct a first-pass filter using simple text-pattern rules (regular expressions) to flag any turn that *might* be referring to what someone else said. I use seven families of cues: (i) a named person plus a speech verb ("Mr. Garcia said . . . "); (ii) a generic description of a speaker ("the gentleman who spoke earlier," "several residents mentioned"); (iii) references to the comment period ("we heard tonight during public comment"); (iv) responsive framing ("to address the concern that was raised"); (v) quotation or paraphrase markers ("as was mentioned," "what I'm hearing is"); (vi) explicit validation ("that's a fair point"); and (vii) tallies of input ("we received forty emails on this"). These rules are deliberately over-inclusive: they flag 25,844 distinct turns (roughly 31,000 turn–item pairs), most of which turn out not to be citations at all. Because one merged speaking turn can contain several flagged turns, the flagged turns fall within 20,852 of the 275,708 merged speaking turns; the complementary 254,856 never-flagged merged turns form the pool for audit (ii) below.

Step 3 (language-model reading): A language model (Gemini 3 Flash) evaluates the flagged turns. To economize, flagged turns first pass a light screen by the same model; the 14,307 turns that survive are then each read in the context of the meeting transcript, and the model answers three questions: Is the councilmember actually relaying something *another person* said (rather than making their own point or running the meeting)? Was that person a member of the public, as opposed to city staff or a fellow councilmember? And was the comment made at *this* meeting, rather than at an earlier hearing or in correspondence? A turn counts toward the headline measure only when the answer to all three is yes—8,672 turns do—and the model records a short paraphrase of the comment being referenced. (Both Step-3 prompts are reproduced below.)

Step 4 (annotation): For counted citations, I conduct two further passes: (a) link the citation to the specific public speaker being referenced and (b) classify what the citation is *doing*, into six purposes. These passes annotate counted citations and do not change the count. The purpose classification (b) is audited in the purpose-validation paragraph at the end of this section; the speaker link (a) is audited where the analyses that use it are presented.

Light-Screen Prompt (Step 3, first pass)

```
You are doing a narrow first-pass filter on short snippets from California city
council meetings. Question: Does this text snippet suggest that the speaker is referring to a
    comment made
by another individual person or group? Return exactly one score for each snippet: - 2 = definitely
```

[9] A "turn" throughout this section is a single diarized segment, the unit in which every pipeline count below is stated. The diarizer splits continuous speech across segments: combined into *merged speaking turns* (same speaker, gaps of at most one minute), the 2.2 million turns entering the pipeline correspond to 275,708 merged speaking turns, about 24 per meeting. Audit (ii) below is conducted in this coarser merged unit.

yes - 1 = maybe / ambiguous - 0 = definitely no. Guidance: - Score 2 when the snippet clearly refers to something someone else said, asked, commented, raised, or communicated. - Score 0 when the speaker is just asking their own question, making their own point, directing procedure, or addressing someone directly without referring back to a prior comment. - Score 1 when the snippet hints at prior comments or public input but is too vague to be confident. - "another individual person or group" includes residents, commenters, staff,
applicants, council members, or broader groups like "communications we've received tonight" when that clearly refers to others' comments.

Examples from prior reviewed outputs: Score 2 examples: 1. "You know, we had the one resident who lived on Black Avenue speak about all the activity" Why: explicitly references a resident's earlier comment. 2. "That's good, because some of the communications we've received tonight are how are we going" Why: explicitly refers to comments/communications received earlier in the meeting. 3. "Ms. Clark mentioned the workforce housing can be a requirement" Why: explicitly refers to what another speaker said.
Score 1 examples: 1. "are sold, but that one unit is retained. Is there a safety mechanism? I share your concern," Why: suggests the speaker may be responding to another person's concern, but the snippet alone is too incomplete to be definitely sure.
Score 0 examples: 1. "Would you like to introduce the one person that is sitting behind you?" Why: direct question, not a reference to a prior comment. 2. "So maybe we can start with the question that is a very general" Why: setting up the speaker's own question, not citing someone else's comment.
Here are the snippets:
<JSON object mapping each turn_id to the flagged turn's text>

Full-Context Reading Prompt (Step 3, second pass)

You are analyzing a California city council meeting transcript that has already
been filtered to discussion on these topics: <list of topic titles used to filter the transcript> The transcript was produced by an automatic speaker diarization system. Speaker IDs like SPEAKER_17 are not names. The diarization can be imperfect, so reason carefully and do not overclaim. You will receive:
1. A filtered meeting transcript, covering only the selected topic discussion. 2. A list of candidate council-member turns that may be referring to comments made by someone else.
For each candidate turn, analyze: 1. Whether the candidate turn is indeed referring to a comment made by another person. 2. If yes, whether that earlier comment was made by a government speaker or a public speaker. 3. If the earlier comment was made by a public speaker, whether it was made in this meeting or in a different meeting. 4. If it was made by a public speaker in this meeting, what the content of the earlier public comment was. 5. If there is a referenced comment, whether its main substance is best characterized as an objective fact claim or a subjective preference/opinion.

Important rules: - Use the filtered meeting transcript as the primary evidence. - If the candidate turn is too vague to support a firm conclusion, say so. - Only mark ‘public_comment_in_meeting‘ true when the evidence in the provided transcript supports that conclusion. Some may be referring to comments made in different meetings, which you are not expected to know about. - If the speaker appears to refer to a public sentiment generally rather than a specific comment by another person, mark it as not a comment citation. - For ‘referenced_comment_content‘, give a concise paraphrase with direct quotations grounded in the transcript. Do NOT invent details. - Keep ‘meeting_summary‘ concise and under 100 words.
Meeting metadata: <town, date, meeting identifier, turn counts>
Candidate turns: <JSON list of candidate turns: turn IDs, speaker ID and diarized name, position, matched cue categories, timestamps, turn text>
Filtered transcript: <the topic-filtered transcript, one line per turn: speaker ID, start/end timestamps, text>

Validation design. The pipeline can *overcount*, by counting turns that are not in fact same-meeting public citations, and it can *undercount*, by discarding true citations at any of the three filters. Four audits, one per error channel and numbered in pipeline order, measure both directions: audit (i) samples the turns discarded by the ten-word floor of Step 1; audit (ii) samples the ten-plus-word turns the Step-2 cue patterns never flagged; audit (iii) measures the precision of the turns Step 3 counted—the overcount channel; and audit (iv) probes false omissions among the turns Step 3 read and rejected. Table 14 collects the resulting error budget.[10]

(i) Below the ten-word floor. The ten-word floor discards 43 percent of raw councilmember turns (1.63 million turns) before any later stage sees them. A turn of fewer than ten words that cites a public comment almost necessarily contains one of the Step-2 cues, so I run the identical Step-2 rules over all discarded turns. They flag 5,619 (0.34 percent); the remaining 1.63 million cueless fragments ("Thank you." "Second." "Aye.") cannot be citations under the definition above and are dismissed. From the 5,619 flagged short turns I drew a random sample of 100 and coded them against the three Step-3 questions. Eight (8.0% [4.1, 15.0]) are citations of a public commenter from that meeting—for example, "Soil samples, somebody brought up soil samples"—while most of the rest are procedural acknowledgments or references to staff and colleagues. Applied to the flagged pool, this implies roughly 450 citation events missed below the ten-word floor [230, 840] (before netting). A same-meeting cross-check nets this down: one of the eight duplicated a citation the pipeline had already counted in an adjacent turn by the same speaker, while the others were distinct events (two occurred in meetings where the pipeline counted no citation at all), implying roughly 390 genuinely missed events, or 4.5 percent of counted volume.

(ii) Outside the cue families. The second undercount channel is a citation in a ten-plus-word turn whose wording matches none of the seven cue families. This audit works in the merged unit: 254,856 of the 275,708 merged speaking turns (92 percent) contain no flagged turn. Because true citations are rare in this pool, I measure its rate with a two-stage design: a high-recall model screen of a stratified random sample of 5,000 never-flagged merged turns (oversampling turns that follow the first public comment on the item, where same-meeting citations must concentrate), followed by a full-context Step-3 reading of every screened positive, anchored by blind human coding of a stratified subsample of the resulting verdicts. The screen flagged 713 turns; the full-context reading confirmed 638, of which 185 (29 percent) were direct exchanges with a commenter at the podium—excluded by definition—leaving 453 candidate deliberative citations. I reviewed a blind, stratified sample of 80 verifier verdicts (40 deliberative confirmations, 20 podium confirmations, 20 rejections). Among the deliberative confirmations I upheld 4 of 40 (10.0% [4.0, 23.1]); among podium exchanges, 0 of 20; among rejections, 0 of 20. Anchoring the funnel to the human rate implies roughly 2,330 uncued citation events [920, 5,380], or 27 percent of counted volume.[11]

(iii) Precision of counted citations. From the 8,672 turns the pipeline counted, I drew a random sample of 130 and labeled an initial random prefix of 48; I agreed with all three calls in 37, or 77.1% [63.5, 86.7]. Of the 11 disagreements, 5 were turns the human judged not to be citations of another person at all, 2 referenced a government speaker rather than the public, and 4 referenced a comment from outside the meeting. Applied to the counted volume, the precision estimate implies roughly 1,990 false positives [1,150, 3,170].

(iv) False omissions among evaluated turns. Of 49 randomly sampled turns that reached the Step-3 reading but were *not* counted, the human coded three (6.1% [2.1, 16.5]) as true same-meeting public citations. Applied to the 5,635 evaluated-but-rejected turns, this implies roughly 340 missed citations [120, 930]. The three misses illustrate the model's main failure modes: a commenter from a nonprofit whose expert-sounding remarks led the model to tag her as a government official; a comment made *tonight about a past event* that the model tagged as coming from a different meeting; and a vague callback ("as was mentioned in the previous item") whose source the model could not place.

Net reading. At the measured rates the pipeline misses roughly 3,070 true citation events across the three undercount channels (dominated by uncued citations) while falsely counting roughly 1,990, a net undercount of about twelve percent of counted volume. Because the four audits are independent samples, their sampling uncertainties can be propagated jointly. Doing so puts the net at +1,200 events, with a 95 percent joint interval of [−850, +4,100], and places 86 percent of the

[10] All intervals are 95% Wilson score intervals, obtained by inverting the two-sided score test for a binomial proportion. The Wilson interval has substantially better coverage than the conventional Wald interval, $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$, when samples are modest or the observed proportion is near zero or one, and it remains within the logical bounds of zero and one. Intervals on counts scale the proportion interval by the relevant pool size.

[11] The interval reflects the binomial uncertainty of the human anchor, the dominant source; the screen and verification rates are estimated on far larger samples. The human-confirmed cases include the characteristic uncued form: attribution separated from the speech verb by an appositive ("Mr. Hanna from the Southwest Regional Carpenters had mentioned . . . "), which evades the named-person-plus-speech-verb pattern.

Table 14: Error budget for the counted citation total (8,672 turns)

Audit	Error channel	Pool (N turns)	Audit (k/n)	Implied citation events: Point est.	Implied citation events: 95% interval
(i)	undercount: discarded by the ten-word floor	5,619	8/100	390	[200, 740]
(ii)	undercount: never flagged by the cue families	254,856	4/40	2,330	[920, 5,380]
(iii)	overcount: false positives among counted turns	8,672	11/48	1,990	[1,150, 3,170]
(iv)	undercount: evaluated but wrongly rejected	5,635	3/49	340	[120, 930]
Net	missed events − false positives			+1,080	[−850, +4,100]

Notes: Each row converts an audited error *rate* into a *count* of citation events; rows follow the audit numbering of the text, which tracks the pipeline steps. "Pool" is the number of councilmember speaking turns exposed to that error channel: the 5,619 turns below the ten-word floor that match a Step-2 cue (i); the 254,856 ten-plus-word merged speaking turns never flagged by any cue family (ii); the 8,672 turns the pipeline counted as citations (iii); and the 5,635 turns the model read at Step 3 but rejected (iv). "Audit" reports the blind human coding of a random sample from that pool: k errors found among n audited turns. Implied events multiply the audited proportion, and its 95% Wilson interval, by the pool size, with two departures from this simple scaling: row (i) nets out same-meeting duplicates of already-counted citations ($8/100 \times 5{,}619 \approx 450$ gross, $\approx$390 distinct events); and in row (ii) the human rate anchors the two-stage screen-then-verify funnel described in the audit-(ii) paragraph rather than scaling the pool directly, so its interval reflects the anchor's binomial uncertainty. Because the four audits are independent samples, the per-row intervals do not sum; the interval on the net row is a joint simulation interval (footnote 12), whose median (+1,200) sits slightly above the plug-in point estimate shown. Rows are rounded to the nearest ten while the net is computed from the unrounded audit arithmetic, so the printed undercount rows sum to 3,060 against an unrounded total of roughly 3,070. Podium exchanges—a member speaking directly with a commenter at the microphone—are excluded from the citation definition throughout (0 of 20 audited podium verdicts were coded as citations).

probability on a net undercount.[12] The headline rate—roughly one in ten public comments is cited in the same meeting by a councilmember—scales with the event count: correcting by the net estimate yields about one in nine; the joint interval bounds the true figure between roughly one in eleven and one in seven. The share of land-use discussions with at least one citation (27.3 percent) moves less than proportionally, because an error shifts it only by flipping an entire discussion, putting the true share near 29–32 percent.[13] In both cases the reported figure is the conservative one.

Citation purposes (Step 4b). Step 4(b) asks what a citation is *doing*—is the councilmember adopting the comment, arguing with it, or merely thanking the speaker? For each citation where the comment-matching pass of Step 4(a) located the specific public comment being referenced (7,987 turns across housing and land-use items—the strongest subset, where both the council turn and the cited comment are observed), the model reads the turn with surrounding context and the matched comment, and returns a short open-ended label for how the comment is being used (prompt below). The model then reads the full inventory of open labels with their frequencies and proposes a canonical taxonomy; each open label is then mapped to its canonical category. I then consolidated by hand the resulting ten canonical labels into six categories, merging labels that differed only grammatically. The six categories are ordered by how strongly the citation binds public input to council action: *Proposal Incorporation* (the comment changes what the council is deciding—a suggestion enters a motion, amendment, or direction to staff; 6.1% of classified citations), *Operational Uptake* (the comment changes what is asked—it prompts a question or clarification request to staff or the applicant; 23.6%), *Policy Justification* (the comment is invoked as evidence or warrant for a position the member is taking; 27.6%), *Rebuttal & Correction* (the comment is disputed or corrected; 9.3%), *Acknowledgment* (participation is recognized—thanks, validation—without operational or evidential consequence; 27.9%), and *Queue & Meeting Management* (administrative handling of speakers; 5.5%, treated as a non-citation residual).

Purpose-Classification Prompt (Step 4b)

```
You are analyzing one council citation case from a California city council meeting. Goal: Infer a broader-
     minded classification for this citation without forcing it into objective fact vs subjective preference.
     You must separately classify: 1. What kind of thing the original public comment contains. 2. How the
     council member is using that public comment. Also decide whether this is: - a substantive citation, - a
```

[12]The joint interval is computed by simulation. Each audited proportion is drawn from its Jeffreys posterior, $\mathrm{Beta}(k+\frac{1}{2},\, n-k+\frac{1}{2})$ with the four audits drawn independently, as their samples are. Each draw is passed through the identical arithmetic that produces the point estimates: the pool scalings for audits (i), (iii), and (iv), the same-meeting duplication adjustment in audit (i) (itself drawn from the posterior of 7 of 8), and, for audit (ii), the stratified never-flagged pool weights and the human anchor of 4 of 40. The net—missed events minus false positives—is formed draw by draw, and the reported interval is the 2.5th–97.5th percentile band over 200,000 draws. The simulation median (+1,200) sits slightly above the plug-in point estimate (+1,080) because the small-sample posteriors are right-skewed, and the share of draws with a positive net (86 percent) is the source of the probability statements in this paragraph.

[13]Of the fourteen genuinely missed events identified across audits (i), (ii), and (iv)—fifteen audited misses less the audit-(i) same-meeting duplicate—six occurred in discussions with no counted citation; the fully-false count applies the 22.9 percent turn-level false-positive rate independently within items over the observed distribution of counted turns per item.

```
     collective citation, - a procedural/courtesy acknowledgment, - or an evidence-gap case where the available
      local evidence is too weak.
Important rules: - Do not force substantive labels when the evidence is weak. - "Evidence gap" means the
     currently provided context does not support a clean reconstruction of the cited comment, even if the turn
     may still be a citation. - "Procedural/courtesy" means the council member is mainly thanking, queueing, or
      generally acknowledging speakers without engaging their content. - Collective summaries are allowed: a
     council member may be citing a shared sentiment or recurring point from several speakers. - Repeated
     questions raised by the public are substantive. Treat them as a real content type and/or council-use type
     when appropriate. - Rhetorical uptake is substantive when the council member reuses a metaphor, flourish,
     slogan, or framing move from the public comment. - The current forward/backward labels are only baselines.
      You may disagree.
Case metadata: <town, date, issue summary>
Council turn with +/-2 context: <the flagged turn and two turns either side>
Citation phrase that triggered the attribution regex: <matched phrase>
Forward-pass interpretation: <Step-3 paraphrase of the referenced comment>
Backward audit interpretation: <matched public comment block and match rationale>
Matched/candidate public block evidence: <text of the matched public comment>
```

Purpose validation. On the same 130-turn random sample of counted citations that precision audit (iii) draws on—each audit labels an initial random prefix of that sample, so the two sets overlap heavily but are not identical: 48 turns enter the precision audit and 49 enter this one, which extends slightly further into the sample and skips turns that duplicate other sampled turns or carry no purpose label to compare—the human coder assigned each counted citation to one of the six purposes, blind to the model's label, reading a wide transcript window (two minutes before the citing turn through four minutes after, plus the matched comment text). The 49 turns labeled, with labeling of the remainder in progress, yield 45 comparable pairs. Six-way agreement is 57.8% [43.3, 71.0] ($\kappa = 0.47$); collapsing the six categories into substantive uptake (the first four) versus ceremonial handling (the last two) raises agreement to 82.2% [68.7, 90.7] ($\kappa = 0.60$).

The disagreements are of two kinds. Eleven of the nineteen are confusions between adjacent categories on the same side of the substantive–ceremonial line, mostly among the substantive purposes: a single citing turn frequently incorporates a suggestion, justifies a position, and rebuts a claim at once, so the forced six-way choice produces near-ties. The remaining eight all run one way: turns the coder read as substantive that the model filed as *Acknowledgment*, with no errors in the reverse direction. The model classifies from the citing turn and the matched comment, and when the substance arrives minutes later—a suggestion enters a motion, staff receive direction—it defaults to the ceremonial label; overall it places 40% of these turns in the two ceremonial categories against the coder's 22%. I therefore treat the six-way split as descriptive, rely on the substantive-versus-ceremonial distinction where the classification enters the analysis, and read the substantive-uptake share as a lower bound and the ceremonial share as an upper bound—the conservative direction for the claim that citations carry operational content.

5 Project Denial Rate

Each voting issue, parsed per the methodology described in Martin and Venugopal (2026), carries a free-text vote outcome recorded from the transcript (e.g., "Appeal denied (ABR decision upheld)"). I classify each outcome with an ordered set of regular expressions: denials of an application, granted appeals of an approval, removals from a housing-site inventory, and withdrawals under a denial motion are coded as denials; approvals, adoptions, consent-calendar passages, and ordinance readings as approvals. Appeals are inverted where necessary: "appeal denied" upholds the underlying approval (an approval), while "appeal granted" overturning an approval is a denial. Applying this to the 26,916 voted housing and land-use items yields 622 denials, a rate of 2.31%. Table 15 shows the full disposition partition.

Table 15: Disposition of voted land-use issues

Outcome bucket	n	%
Approved	22,521	83.7
Continued	1,588	5.9
Direction to staff	938	3.5
Denied	**622**	**2.31**
Other (unmatched)	766	2.8
Referred	310	1.2
Withdrawn	171	0.6
Total	26,916	100.0

Denial rates by entitlement type. A 2.3% denial rate over all voted items understates the rate where denial is a real possibility, because the denominator is dominated by routine, largely ministerial actions—consent-calendar items, program and contract approvals, technical map approvals, second readings; housing-services items are particularly routine, with a denial rate near 1.4%. To isolate discretionary entitlements, I flag five instruments from the issue title and summary using high-precision keyword patterns (listed in the notes to Table 16). Denials concentrate sharply in quasi-judicial, project-specific permits: appeals, variances, and conditional use permits are denied at 8–12%, several times the overall rate, whereas legislative-style instruments (rezones, general-plan amendments) are denied at council in under 3% of cases. Even pooling all discretionary instruments, the denial rate is 5.4%.

Table 16: Denial rates by discretionary-entitlement instrument

Subset	Denied	N	Rate (%)
Appeals	188	1,626	11.6
Variances	9	87	10.3
Conditional use permits	52	643	8.1
Rezones / GP amendments	69	2,491	2.8
Subdivisions / maps	22	923	2.4
Any discretionary instrument	275	5,082	5.4
Routine / ministerial	347	21,834	1.6
All voted items	**622**	**26,916**	**2.31**

Notes: Instruments are case-insensitive keyword flags matched on issue title + summary and may overlap (e.g. an appeal of a use permit): appeal (`\bappeal`); variance (`\bvariance`); CUP (`conditional use`, `use permit`, `\bcup\b`); rezone/GPA (`rezon`, `zone change`, `zoning amendment`, `general plan amendment`, `\bgpa\b`, `prezon`); subdivision (`subdivision`, `tentative tract/parcel map`, `tract map`, `parcel map`).

Validation against official minutes. The denial labels rest on outcomes the model read from meeting transcripts. To check for false positives and false negatives, I use the official written minutes that city clerks publish after each meeting, which record what was actually decided. Where cities post these documents in machine-readable form, I downloaded them: 3,216 of the 26,916 items (12%) across 1,568 meetings in 25 cities, including 95 of the 622 flagged denials. Coverage skews toward mid-size cities with modern document portals.

False positives. For 91 of the 95 denials with minutes (96%), the minutes contain explicit denial language for that meeting. The remaining 4 are meetings whose minutes simply do not record outcomes (agenda-style or summary documents), so they are uninformative rather than contradictory. In no case do the minutes affirmatively contradict a flagged denial.

False negatives. We then searched every line of every covered minutes document for any language that could indicate a denial—"denied," "rejected," "motion failed," "appeal granted," and variants—deliberately casting a wide net. The search returned 1,507 candidate lines. After setting aside failed amendments, bid rejections, and boilerplate, 645 lines with strong denial language in a voting context remained as candidates for a missed denial. I hand-checked a random sample of 50 of these, reading each line in context and judging whether it shows a land-use or housing project being denied. The review found 0 missed denials: every candidate turned out to concern something else—failed procedural motions, appointments to boards, contracts and leases, license items—not a project being turned down. Table 17 translates this into worst-case terms: with 0 misses in 50 random draws, the exact one-sided 95% upper bound allows up to 38 missed denials among the 645 candidates, meaning the pipeline caught at least 71% of true denials in the worst case, and the true denial rate is at most 3.2% against our estimate of 2.3%.

Table 17: Worst-case bound on missed denials from the hand-checked minutes sample.

Candidate lines with denial language	645
Candidates hand-checked (random order)	50
Missed denials found	0
Maximum missed denials (95% Clopper-Pearson upper bound)	38
Worst-case share of true denials caught	71%
Worst-case denial rate (vs. 2.3% estimate)	3.2%

Notes: The Clopper-Pearson bound tells you the largest p–or true fraction of the 645 that are missed denials–that is plausibly consistent with seeing 0 misses in 50 draws 5% of the time (solving for Pr(0 misses in 50) = $(1-\bar{p})^{50} = 0.05$ provides $\bar{p} = 0.058$ and $0.058 \times 645 \simeq 38$. "Worst-case share of true denials caught" compares the number of denials caught (95) to the total of those caught plus potentially missed (95+38 = 133). The worst-case denial rate rescales the headline 2.3% by the inverse of the share of true denials actually caught, or 2.3% $\times \frac{133}{95}$.

6 Participation and Project Delay: SF Planning Commission Exercise

Data. The proposal-level data are assembled from the San Francisco Planning Commission records—agendas, minutes, and staff-report packets for 1,103 meetings from January 1998 through March 2026—parsed into a hearing-by-hearing history of appearances, continuances, dispositions, votes, and 45,914 stance-coded speaker appearances across 9,137 proposals (8,459 with a Commission case; 1,164 with five or more residential units) following the methods of Sahn (2025) and extended to the most recent years. New to this paper, I also scrape the staff-report packets the Commission has posted with each agenda since late 2015—94 percent of five-plus-unit proposals heard from 2016 on have one—and extract from them the project characteristics, substance flags, and parcel identifiers used below.

Table 18 compares this source universe with the Annual Progress Reports (APRs) filed with the state housing department, and the Board of Supervisors chains in this paper's council corpus. The APR is a census of housing applications but records outcomes only and 64 percent of its projects are one-to-four-unit buildings, few of which ever reach the Commission. The Board sees a project only when it generates Board business—a rezoning, an appeal, city financing—which covers about one in ten of the Commission's five-plus-unit caseload. The Commission is where San Francisco's discretionary approvals are made. Of 287 project-level Board chains observed in the transcript corpus, 145 match a Commission case, and among the 14 that appear in the APR without one, every residential site of five or more units was entitled ministerially under SB 35 or through the Mission Bay redevelopment authority—housing with no discretionary process to record.

Table 18: Comparative views of San Francisco's housing pipeline

	Planning Commission	Board of Supervisors (transcript corpus)	HCD Annual Progress Reports
Underlying records	Meeting agendas, minutes, staff-report packets	Meeting transcripts, linked into per-project issue chains	Jurisdiction's statutory annual filing
Window	Jan. 1998–Mar. 2026 (1,103 meetings)	Jan. 2015–Mar. 2024	Reporting years 2018–2025
Unit; count	Proposal (site × planning case): 9,137, of which 8,459 with a Commission case and 7,283 on ≥1 agenda	Issue chain: 764 housing-related, 287 project-level	Housing project: 1,051
Enters the data when...	the project requires a discretionary Commission action (conditional use, large-project or downtown authorization, discretionary review, CEQA certification)	the project generates Board business (rezoning or development agreement, appeal, city financing, call-up)	a housing development application is filed—any size, any pathway, ministerial included
Process observed	Hearings, continuances, votes; each public commenter with clerk-coded stance	Board hearings and votes; full-text comment	None
Outcomes observed	Commission disposition	Board disposition	Entitlement, permit, and completion dates; unit counts
≥5-unit projects seen per year (2018–23)	≈38 first heard at a Commission hearing	≈7 entering a Board project chain	≈50 applications filed

Sample and specification. The estimation sample is proposals of five or more residential units, heard through 2024, with complete covariates ($N = 581$).[14] For proposal i in neighborhood n first heard in year t, I estimate by OLS:

$$y_i \;=\; \beta_S \log(1 + S_i) \;+\; \beta_O \log(1 + O_i) \;+\; X_i'\gamma \;+\; \delta_n \;+\; \tau_t \;+\; \varepsilon_i, \qquad (2)$$

where S_i and O_i count supporting and opposed speakers at the proposal's first observed Commission hearing—measured before any continuance, so public comment cannot accumulate mechanically as an item returns to the calendar. X_i contains log proposed units and a mixed-use indicator; five repeat-player flags (land-use counsel or permit expediter, major architect, major general contractor, owner–builder, and Reuben, Junius & Rose); six staff-report substance flags (residential demolition, tenant displacement, individually historic resource, design-review concerns, loss of a community-serving use, and variance or exception requests) plus an indicator for staff-report availability; and log tract median household income and log population density (ACS 2020). δ_n and τ_t are neighborhood (20) and hearing-year (26) fixed effects.

Outcomes and results. Two outcomes capture procedural delay. The first counts continuances—each one a Commission vote to put the item off to a later meeting rather than decide it. The second is the proposal's total time before the Commission, first hearing to last. Delay is concentrated in a long tail: 52 percent of proposals are never continued, and the median first-to-last span is seven days. But proposals continued at least once typically spend two months on the calendar (median 63 days), each further continuance corresponds to roughly another month, and the sample-mean span is 185 days.

Table 19 reports the estimates. To fix magnitudes, compare two otherwise-identical proposals, one facing a single opposed speaker at its first hearing and the other none.[15] The opposed proposal is continued 0.67 more times—40 percent above the sample mean of 1.6—and spends about 70 percent longer before the Commission; for a proposal on the two-month track

[14] Sample construction, from the Commission column of Table 18:

	Proposals
Five-plus-unit proposals with a Commission case	1,164
heard at ≥1 Commission hearing	1,004
first heard through 2024	995
with a neighborhood assignment	685
with a census-tract match (estimation sample)	581

No observation is lost to the speaker counts or staff-report fields; staff-report absence is absorbed by the availability indicator. Because a proposal is a site–case pair, the 581 proposals sit on 526 distinct sites.

[15] The participation regressors enter as $\log(1 + \text{count})$, so going from zero speakers to one is the largest step on the scale, $\log 2 \approx 0.69$. The quantities in the text multiply each coefficient by $\log 2$; the time-span conversion is $e^{0.776\times\log 2} - 1 \approx 71$ percent.

typical of continued items, that is roughly six extra weeks. A first supporting speaker cuts the other way, slightly harder: 0.77 fewer continuances and a time span roughly halved. These are conditional correlations, not causal effects; opposition could still respond to project features the controls miss.

Table 19: First-hearing participation and SFPC procedural delay

	(1) Continued meetings	(2) Commission time span, log(1 + days)
log(1 + supporting speakers)	−1.113***	−1.081***
	(0.208)	(0.188)
log(1 + opposed speakers)	0.965***	0.776***
	(0.302)	(0.223)
Project controls	✓	✓
Repeat-player controls	✓	✓
Staff-report substance controls	✓	✓
Tract income and density	✓	✓
Neighborhood fixed effects (20)	✓	✓
Hearing-year fixed effects (26)	✓	✓
Mean of dependent variable	1.61	2.36
Observations	581	581
R^2	0.209	0.297

Notes: The unit of observation is a San Francisco Planning Commission proposal of five or more residential units, heard through 2024. Both columns estimate equation (2) by OLS with HC1 robust standard errors in parentheses. Continued meetings is the number of times the item was continued to a later Commission meeting (sample mean 1.61); commission time span is $\log(1 + \text{days})$ from the proposal's first to last Commission appearance (median 7 days, mean 185). Supporting and opposed speaker counts are measured at the first observed Commission hearing. Controls and fixed effects are described in the text; coefficients are absorbed, not printed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Project Chains and Conditions

7.1 Construction of Multi-Meeting Project Chains

A development project often appears before a council many times. For example, a large project can surface as a study session, an environmental-review certification, the entitlement hearings themselves, and later amendments, spread over years. Because conditions of approval accumulate across these appearances, the unit of analysis is the project, not the agenda item, and land-use agenda items must first be linked into project chains. The base linkage joins items within a city when they share a concrete project anchor (a street address or parcel number) or when an LLM pair-matcher judges two candidate items to concern the same underlying matter.

This base linkage leaves projects badly fragmented—81 percent of its 21,303 chains contain a single item, and Tracy Hills, a 5,500-unit master-planned community in Tracy with 66 agenda items across more than thirty meetings from 2016 to 2023, was scattered over sixteen separate chains. Fragmentation is worst for exactly the multi-year projects where negotiated conditions concentrate, so I layered three merge passes on top. The first derives a deterministic fingerprint for each item from its title—in priority order, a street address, an assessor's parcel number, the target of an appeal, or a distinctive title prefix—and merges items with the same fingerprint in the same city, rejecting prefixes that appear in three or more cities (those are policy themes, not projects). The second merges chains that share two or more rare project-name tokens within a three-year window, or a normalized street address. The third has an LLM (Gemini 3 Flash) assign each item a canonical project name and a project-versus-citywide-policy label; same-named items in the same city are merged, and clusters that turn out to be citywide programs rather than projects are split back apart.

Together the passes cut the chain count from 21,303 to 19,311 and the single-item share from 81 to 76 percent, and reunify the known hard cases: Tracy Hills's sixteen fragments, the North 40 project in Los Gatos (25 fragments), and Alameda Point (14) each collapse to a single chain. Spot-checking the remaining single-item chains shows they are mostly genuine one-touch items—routine variances, consent-calendar approvals, single-hearing use permits. These 19,311 chains are the project histories the main text describes. Most are not development projects: the third pass's labels anchor 7,745 chains to a concrete project (a specific address, parcel, or named project), of which 7,707 carry an unambiguous project type—2,394 spanning multiple meetings and 5,313 containing a single item. The condition extraction described next reads the 2,394 multi-meeting project chains and the 3,408 single-item project chains with more than five minutes of hearing time—5,802 chains in all, submitted as 7,617 constituent records—which keeps citywide programs and ordinances out of the project universe and skips one-touch items too brief to carry conditions. Table 20 tracks every count from the base linkage to the estimation samples.

Table 20: Project chains and conditions: sample flow

Panel A. Project-chain construction and verification	
Chains from the base linkage	21,303
After the three merge passes	19,311
Concrete projects with an unambiguous type	7,707
Read by the condition extraction (2,394 multi-item; 3,408 eligible single-item)	5,802
In-scope private developments	4,232
Yielding ≥1 verified condition	3,749
Panel B. Condition rows	
Raw extracted condition rows	18,951
Unique candidate conditions	17,809
Verified conditions	12,193
Panel C. Multi-meeting residential analysis sample	
Multi-meeting project chains	2,394
Residential project chains	1,734
In-scope analysis chains	1,521
First-comment projects	611
Complete-covariate projects	491

Notes: The 7,707 concrete projects exclude 11,566 chains not anchored to a specific project and 38 without an unambiguous project type. The condition extraction excludes single-item chains with five minutes of hearing or less. Out-of-scope chains include city programs, the city's own transactions, rent or tenant matters, and paper land actions. Candidate conditions are rejected when they occur on an out-of-scope chain, do not represent an adopted condition, or bind the city or a third party rather than the applicant. The 12,193 verified conditions comprise 8,725 negotiated, 3,464 mandatory, and 4 unclear conditions. "Multi-meeting" means at least two agenda items; 37 of the 1,734 residential chains appear more than once within a single meeting.

7.2 Annotation of Project Conditions

Conditions of approval are extracted from the 5,802 project chains in Panel A: all 2,394 multi-item chains and the 3,408 single-item chains with more than five minutes of hearing time. For each chain, an LLM reads the full transcript record and the chain's stage history and returns each condition with a category, a short description, and a mandatory-vs.-negotiated call. The conditions block of the extraction prompt is below (the same call also extracts project attributes and between-meeting changes; those blocks are omitted here).

Extraction Prompt (conditions block)

```
You are extracting land-use project information from a chain of city council meeting records about a single
     project. [...] Extract what the CONTEXT ACTUALLY SHOWS. If an attribute value is not stated, OMIT IT -- do
      not guess. Every evidence_quote must be a near-verbatim substring of the provided context.
CONDITIONS OF APPROVAL (per stage: the STOCK, not just deltas). For every stage, emit the full list of
     conditions present at that stage. If carried forward, emit again. Categorize each from this set:
     design_aesthetic, operational_hours, use_restriction, access_circulation, landscaping_buffer,
     affordability_covenant, tenant_protections, community_benefit, construction_management,
     environmental_monitoring, phasing_timeline, public_art, noise_lighting, tree_preservation,
     parking_management, traffic_mitigation, other.
For each condition fill: description (<=200 chars); binding: condition_of_approval | advisory |
     voluntary_commitment; imposed_by: council_motion | staff_report | settlement | state_mandate |
     applicant_proffer | planning_commission | design_review_board | local_ordinance (standard CUP / municipal-
     code recital) | prior_entitlement (carried forward from prior approval, no new negotiation); bucket:
     mandatory | negotiated | unclear (mandatory = required by statute, court, or state mandate, no real
     negotiation; negotiated = a discretionary ask added by council/staff/applicant; unclear = boilerplate
```

```
where the line is genuinely fuzzy); bucket_reasoning; dollar_amount_usd when a specific dollar value is
     tied to this condition; evidence_stage_idx where the condition was first imposed.
```

De-duplicating restatements of the same condition across a chain's hearings leaves 17,809 unique candidate conditions across 5,247 project chains. My main concern with this output was over-extraction: hand-reading early samples showed that alongside true conditions it captured items from chains that are not development projects at all (city programs, the city's own land deals, rent-board matters), project features and unadopted public demands described in the same breath as conditions, and obligations that bind the city rather than the applicant. I therefore ran a second verification pass over every extracted condition:

Verification Prompt

```
You audit "conditions of approval" that an earlier model extracted from California city-council land-use
     hearing transcripts. You receive one CHAIN (all hearings on one project/matter): its name(s), city, dates,
      chain-wide STAGE SUMMARIES, and a numbered list of extracted conditions, each with a transcript excerpt.
     Return STRICT JSON [...]
CHAIN SCOPE. in_scope=true only for private development / entitlement matters -- a private (or nonprofit/public
     -private) applicant seeking approval to build, expand, convert, subdivide-for-development, or entitle a
     specific project. Affordable-housing DEVELOPMENT projects are in scope. NOT in scope:
     city_program_or_agreement (city service programs, operating MOUs/licenses with nonprofits, e.g. a shelter
     OPERATIONS agreement); city_own_transaction (the city buying/selling/exchanging its own property);
     rent_or_tenant_matter; paper_land_action (lot mergers/subdivisions "for conveyance purposes only");
     other_non_development.
cond_type (judge each extracted condition from the excerpt AND the stage summaries): condition = an obligation
     BINDING ON THIS PROJECT's approval -- imposed by council/staff/commission, adopted into a motion or
     development agreement, OR required by code/ordinance/CEQA and applied to this project. Adopted community-
     benefit or land-dedication packages count as condition (negotiated), even when the developer nominally "
     offered" them. applicant_offer = a voluntary proffer NOT (yet) adopted into the approval. project_feature
     = an attribute of the project as proposed; nothing imposed or offered in exchange for approval. When torn
     between applicant_offer and project_feature, choose project_feature. demand_only = someone asked for it
     and it was NOT adopted. Before choosing this, CHECK THE STAGE SUMMARIES for adoption at a later hearing;
     if adopted later, it is a condition. discussed_only = described in the abstract (what a zone permits
     generally, another project's terms, a contingent future process); no obligation attached. relief_granted =
      a waiver/variance/reduction granted TO the applicant. other.
IMPORTANT PRESUMPTION: your excerpt is a PARTIAL window; the extractor saw the full record. Hearings routinely
     adopt the staff-recommended conditions in one final motion that may fall outside your excerpt. If the
     excerpt shows the item being stated as a requirement/condition FOR THIS PROJECT, classify it as "condition
     " -- do NOT demote to discussed_only or demand_only merely because you cannot see the adoption vote in the
      excerpt.
bound_party: who bears the obligation -- "applicant" (developer/property owner), "city" (the city itself pays
     or acts), "third_party", or "unclear".
negotiated: "mandatory" = standard/code-required/CEQA-required, applies to every such project; "negotiated" =
     bargained, discretionary, or extracted/offered in exchange for approval; "unclear" if the record doesn't
     say.
Judge only from the provided text. Do not guess what the extractor "probably" meant.
```

A condition enters the final dataset if its chain is a private development matter and the item is a condition or adopted applicant offer binding the applicant. This retains 12,193 of the 17,809 candidates (68.5%) on 4,232 of the 5,247 chains, of which 8,725 are classified as negotiated and 3,464 as statutorily mandated.

Validation. I hand-labeled condition rows in three rounds, blind to the models' labels in each. The first two rounds (40 rows each, drawn from the raw extraction) are what motivated and shaped the verification pass—only 58% of the first sample were real conditions, and my margin notes on those rows became the verification prompt's scope and type rules (the second sample, under the revised protocol, came in at 80%). The reported validation is a third, held-out sample drawn after the verification pass: 40 fresh rows from chains I had not previously seen, half from conditions the verifier retained and half from conditions it dropped, with the strata hidden and rows in random order. I labeled each row from the condition text and a transcript window, revisiting disagreements against wider windows before finalizing.[16]

[16]Two protocol notes. First, the labeling form initially lacked an "out of scope" option, which forced a handful of city-program rows into condition-shaped answers; I re-ruled those once the option was added. Second, several disagreement rows were discussed with a higher reasoning model–Claude Fable 5–against the Gemini model's verdicts before my final pass; I ultimately sided against the model on about half of them, so the final labels are not simply anchored to the model.

Of the 20 retained conditions, I agreed with 19 (95%; 95% CI [76, 99]); the one exception was a row where the extractor had recorded a project's $45 million land purchase price as if it were a condition. Of the 20 dropped rows, I judged 3 (15%, [5, 36]) to be real conditions the verifier should have kept—all three are obligations whose adoption language is scattered across the record. The audit deliberately sampled equal numbers of retained and dropped rows, whereas the full candidate pool is 68.5% retained and 31.5% dropped. Reweighting the two audit strata to match the population implies that the verifier retained approximately 93% of the real conditions and agreed with my labels on approximately 92% of all candidate rows. In other words, the final dataset omits some genuine conditions, but contains relatively few apparent nonconditions.

Project size changes. The same extraction produces a candidate change log for quantitative project attributes, including unit count, building height, parking, and affordable units. For each purported change, the model records the before and after values. The raw log contains 752 candidate quantitative changes, of which 484 concern unit counts. A 39-row human calibration audit found that only about half represented genuine changes in a project's total number of dwelling units. I therefore screen every unit-count entry before using it. This leaves 404 candidate increases or decreases. An LLM verification step then checks whether both numbers refer to the same project's total dwelling-unit count and whether the transcript describes an actual revision; 256 claims, or 63 percent, pass this screen. I refer to these as screen-passing rather than human-verified changes. They occur in 241 projects and are nearly evenly divided between decreases and increases (140 versus 116). I recompute magnitudes from the retained before and after values: the median decrease is 12 percent and the median increase is 17 percent. The excluded claims often compare a proposal with zoning capacity, an alternative under discussion, or a single phase with the entire project, which helps explain why changes in the raw log appear larger.

Validation of the unit-count change log. To validate the screen, I hand-labeled a stratified sample of 39 raw entries, judging from the transcripts whether each logged change was a real revision of the project's unit count. Only 51 percent [36, 66] of raw entries were—which is why the screen is needed. The screen's keep-or-drop decisions agree with my labels 87 percent of the time [73, 94]: rows it keeps are genuine changes 85 percent of the time [64, 95], and it retains 89 percent [69, 97] of the genuine changes.

7.3 Participation and Project Conditions

Data. The unit of analysis is a development project followed across its city council appearances, documenetd in the data as land-use agenda items linked across meetings into 1,734 multi-meeting project chains.[17] The verified set of 12,193 conditions is the outcome measure throughout this section. Participation is measured at the project's first observed hearing with public comment, restricted to the public: nongovernmental speakers, excluding business- and applicant-affiliated ones.[18] Comment stance comes from a per-comment attitude score described earlier (opposed below −0.5, supportive above +0.5); sample projects draw a mean of 2.9 opposed and 2.1 supporting speakers at that hearing.

Level margin. The first question is whether participation predicts how many conditions attach. The estimation sample is the 611 in-scope project chains with at least one public comment at a first hearing, of which 491, in 91 cities and first heard 2015–2024, have complete covariates. For project j in city c first receiving comment in year t, I estimate by OLS:

$$\log(1 + C_j) \;=\; \beta_S \log(1 + S_j) \;+\; \beta_O \log(1 + O_j) \;+\; X_j'\gamma \;+\; \delta_c \;+\; \tau_t \;+\; \varepsilon_j, \tag{3}$$

where C_j counts the project's verified conditions and S_j and O_j count supporting and opposed speakers at the first hearing with comment. X_j contains log proposed units, an affordability indicator and the affordable share, and project-type indicators; log tract median household income, log tract median home value, and housing density; and two record-length controls—log agenda items and log meetings in the chain—so longer project records cannot mechanically carry both more recorded comment and more extracted conditions. δ_c and τ_t are city (91) and first-comment-year (10) fixed effects; standard errors are clustered by city.

92 percent of sample projects carry at least one verified condition, with a mean of 4.2, leaving little extensive-margin variance for participation to explain. Panel A of Table 21 reports the estimates: the supporting- and -opposed-speaker coefficients are small, statistically indistinguishable from zero, and indistinguishable from each other in every column. Splitting conditions into negotiated terms and mandatory code requirements changes nothing, and extensive-margin linear probability models are equally flat.

[17] I first attempted condition extraction in the San Francisco Planning Commission records I used for the delay. Conditions in these records are observable only in the staff packet as a standardized "Exhibit A." However, these were almost entirely boilerplate recitals that seemed to bear no relation to commenters, and the minutes proved too sparse to tie conditions to comments. The council transcripts, on the other hand, capture conditions as negotiated in the room and the full text of every comment and add cross-city variation.

[18] The restriction matters: under the raw "public comment" flag, half of supportive speech at these hearings comes from council members and staff, and a further third of the nongovernmental remainder from the applicant's own team.

Composition margin. What participation does predict is *which* conditions attach. I classify each of the 3,491 public comments at first-comment hearings, multi-label, into the sixteen substantive categories used for conditions (the extraction prompt's seventeenth label, the residual `other`, is excluded on both the comment and condition sides of the panel), and estimate, for project j and condition category c,

$$\text{Condition}_{jc} = \beta \,\text{Raised}_{jc} + \alpha_j + \gamma_c + \varepsilon_{jc}, \tag{4}$$

where $\text{Condition}_{jc} = 1$ if project j's record contains a verified condition of category c, and $\text{Raised}_{jc} = 1$ if a member of the public raised category c at project j's first hearing with comment. Each project contributes sixteen observations, one per condition category, for a panel of $611 \times 16 = 9{,}776$ cells. The project effects α_j give each project its own intercept, so anything constant about the project (city, size, year, how contested it was, how many conditions it drew in total, how many people spoke) cannot move the estimate, which is why the specification carries no additional controls. The category effects γ_c give each category its own baseline, so the estimate is not inflated by categories that are common on both sides of the equation everywhere. β is identified by the comparison, within a single project, of the categories the public raised against the categories it did not. Standard errors are clustered by city (96).

Panel B of Table 21 reports the estimates. A category raised at the first hearing is 9.4 percentage points more likely to appear among the project's verified conditions than a category no commenter raised, against a base rate of 15.9 percent—roughly 60 percent above baseline. The estimate is robust to replacing γ_c with city $\times$ category fixed effects and to restricting the panel to the level-regression sample (+9.5 percentage points, SE 1.4), and it is positive and significant for 11 of 16 categories individually.

Two further checks establish what this correlation is, and what it is not. First, it is not an artifact of the condition-verification pass. One might worry that the pattern is mechanical—that the extraction or verification step is more likely to record a condition when nearby comment text mentions the same subject. But the same regression run on the raw extracted conditions, before any verification, yields a proportionally similar estimate (+10.5 points on an 18.7 percent base), so verification did not create the pattern; and run on the candidate conditions the verification *rejected*—where extraction noise concentrates—the relationship is proportionally weaker (+2.5 points on a 4.9 percent base). Second, the match is category-specific. Comparing raw means category by category, conditions of a category are on average 1.72 times as likely to appear when commenters raise that category as the category's unconditional base rate, while conditions in *other* categories barely move (mean lift 1.02).[19] Projects that draw comment do not simply accumulate conditions of every kind; the condition that appears is the one on the subject commenters raised.

[19] Both lifts average cells of the 16×16 matrix of raised-category by condition-category raw conditional means over unconditional category means—the diagonal (1.72) and the off-diagonal (1.02). Because this comparison equal-weights the sixteen categories and adjusts for nothing, the diagonal average is larger than the pooled lift implied by the regression estimate, $(15.9 + 9.4)/15.9 \approx 1.6$.

Table 21: First-hearing participation and approval conditions: level and composition

	(1) All conditions	(2) Negotiated only	(3) Mandatory only
Panel A. Level: project level, $\log(1 + \textit{conditions})$			
$\log(1 +$ supporting speakers$)$	0.017	0.017	−0.015
	(0.048)	(0.055)	(0.040)
$\log(1 +$ opposed speakers$)$	0.017	0.017	0.015
	(0.046)	(0.041)	(0.050)
Mean of dependent variable	1.42	1.18	0.53
Observations (projects)	491	491	491
R^2	0.409	0.414	0.298
Panel B. Composition: project × category level, any condition in category			
Category raised at first hearing	0.094***	0.067***	0.034***
	(0.012)	(0.010)	(0.007)
. . . raised in opposition	0.069***	0.054***	0.023**
	(0.015)	(0.014)	(0.009)
. . . raised in support	0.086***	0.064***	0.026**
	(0.017)	(0.014)	(0.011)
Mean of dependent variable	0.159	0.121	0.048
Observations (project × category)	9,776	9,776	9,776
R^2	0.187	0.173	0.129

Notes: Column (1) uses all verified conditions, and columns (2)–(3) split by negotiated/mandatory label. Panel A estimates equation (3) by OLS on the 491 in-scope project chains with a public comment at the first hearing with comment and complete covariates; the dependent variable is $\log(1 + \text{conditions})$ anywhere in the project's record, and project, record-length, and tract controls plus city (91) and first-comment-year (10) fixed effects are absorbed, not printed. Extensive-margin LPMs are equally flat: any negotiated (base 0.841), supporting 0.030 (0.025), opposed 0.012 (0.024); any mandatory (base 0.544), supporting 0.018 (0.036), opposed −0.015 (0.048); any condition (base 0.923), supporting 0.036 (0.019), opposed −0.000 (0.021)—the largest of these, supportive speakers on the probability of any condition, is significant only at the ten-percent level. Panel B estimates equation (4) as a linear probability model on the 611 × 16 project–category panel with project and condition-category fixed effects; the dependent variable equals one if the project's record contains a verified condition of the category, and *category raised* equals one if a public commenter raised it at the first hearing with comment. The top row enters the pooled indicator; the opposition/support rows are a separate specification entering both jointly; their equality is never rejected ($p \geq 0.50$). Standard errors clustered by city in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At least two readings of this pattern are possible. The first is responsiveness: the council hears the concern and writes the condition. In Menifee, for example, neighbors on one-acre horse properties objected to two-story homes overlooking their lots, and the approval added an eight-foot wall and a one-story limit on the facing street. The second is shared observation: a single site attribute can prompt both the complaint and the condition with no transmission between them—the adjacency that brought the Menifee neighbors to the microphone is the same adjacency a planner reviewing the site plan might have conditioned on regardless, and because such an attribute varies across categories *within* a project, the fixed effects in equation (4) do not rule it out. To gauge the prevalence of the second reading, I audit two hundred matched cases—a condition in a category the public raised, recorded at the hearing where they raised it—and use the diarized, timestamped transcript to locate where in the hearing the condition's substance first appears.[20] The result is an almost even split. In half (45.7 percent), the condition appears first in the staff presentation, before any member of the public has spoken. In the other half (45.7 percent), the condition is first articulated only *after* the public comments—in a council member's motion, a staff revision, or a concession the applicant offers in the room (6.7 percent are not locatable in the excerpt). Only two percent (1.8) of audited conditions exist solely in public comment, consistent with the verification pass, which independently types just 2.8 percent of extracted candidates as unadopted podium demands and, in the verified set used here, removes them by construction. Either reading supports the conclusion that the public hearing operates as an exchange over a project's terms.

[20]The audit stays within the hearing deliberately: most condition packages originate below the council, in staff review and at the planning commission, which these transcripts do not observe, so timing *across* hearings cannot separate the two readings. Each audit verdict carries a timestamped near-verbatim quote.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023a. doi: 10.1126/science.adi6000.

Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. PPI++: Efficient Prediction-Powered Inference, 2023b. URL `https://arxiv.org/abs/2311.01453`. arXiv:2311.01453.

Cameron Blevins and Lincoln Mullen. Jane, John ... Leslie? A Historical Method for Algorithmic Gender Prediction. *Digital Humanities Quarterly*, 9(3), 2015. URL `http://www.digitalhumanities.org/dhq/vol/9/3/000223/000223.html`.

Katherine Levine Einstein, David M. Glick, and Maxwell Palmer. Neighborhood Defenders: Participatory Politics and America's Housing Crisis. *Political Science Quarterly*, 135(2):281–312, 2020. doi: 10.1002/polq.13035.

Marc N. Elliott, Peter A. Morrison, Allen Fremont, Daniel F. McCaffrey, Philip Pantoja, and Nicole Lurie. Using the Census Bureau's surname list to improve estimates of race/ethnicity and associated disparities. *Health Services and Outcomes Research Methodology*, 9(2):69–83, 2009. doi: 10.1007/s10742-009-0047-1.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic Forecasts, Calibration and Sharpness. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Kosuke Imai and Kabir Khanna. Improving Ecological Inference by Predicting Individual Ethnicity from Voter Registration Records. *Political Analysis*, 24(2):263–272, 2016. doi: 10.1093/pan/mpw001.

Olivia Martin and Amar Venugopal. Participation and Representation in Local Government Speech, 2026. URL `https://arxiv.org/abs/2604.21202`. arXiv:2604.21202.

Alexander Sahn. Public Comment and Public Policy. *American Journal of Political Science*, 69(2):685–700, 2025. doi: 10.1111/ajps.12900.

Gaurav Sood and Suriyan Laohaprapanon. Predicting Race and Ethnicity From the Sequence of Characters in a Name, 2018. URL `https://arxiv.org/abs/1805.02109v1`. arXiv:1805.02109v1.

Faiz Surani, Lindsey A. Gailmard, Allison Casasola, Varun Magesh, Emily J. Robitschek, and Daniel E. Ho. What Is the Law? A System for Statutory Research (STARA) with Large Language Models. In *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL '25)*, pages 394–398. Association for Computing Machinery, 2025. doi: 10.1145/3769126.3769219. URL `https://dho.stanford.edu/wp-content/uploads/STARA.pdf`.